CE 490 – Introduction to Digital Image Processing

Image Enhancement – Filtering in Frequency Domain

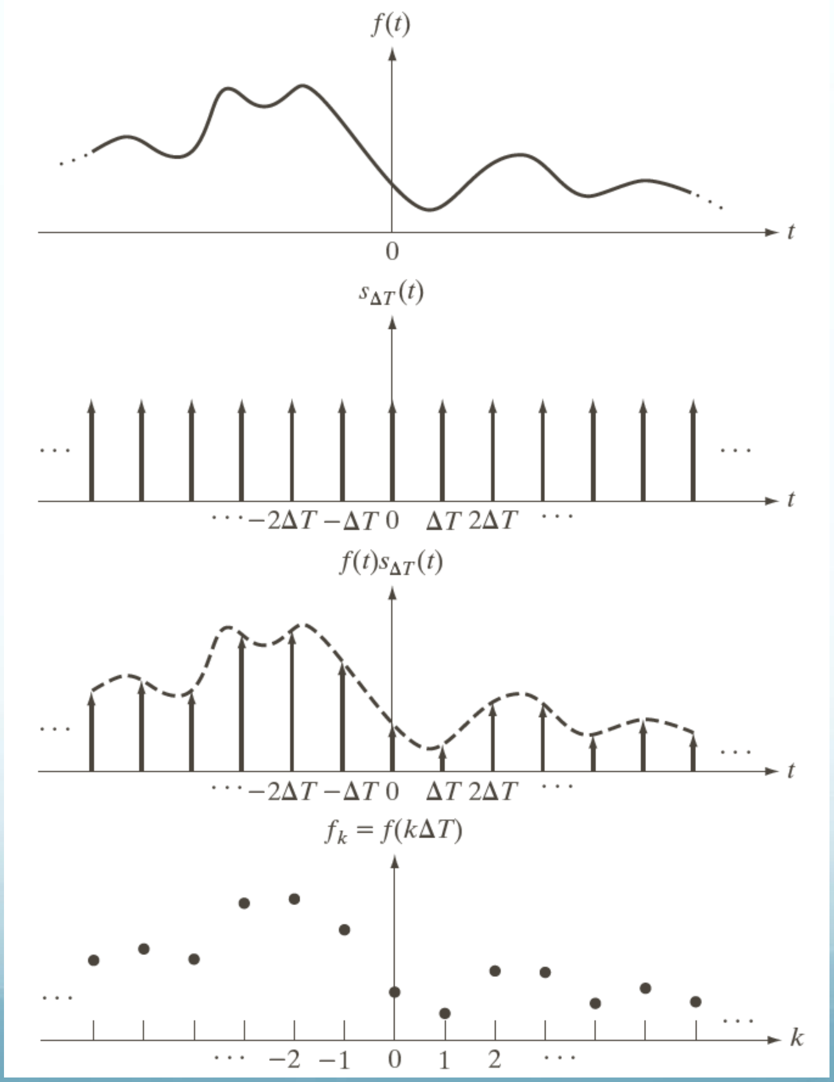
Contents

Sampling, aliasing

Then, we will look at image enhancement in the frequency domain

- Jean Baptiste Joseph Fourier
- The Fourier series & the Fourier transform
- Image Processing in the frequency domain
  - Image smoothing
  - Image sharpening
- Fast Fourier Transform (FFT)

Sampling – 1D example

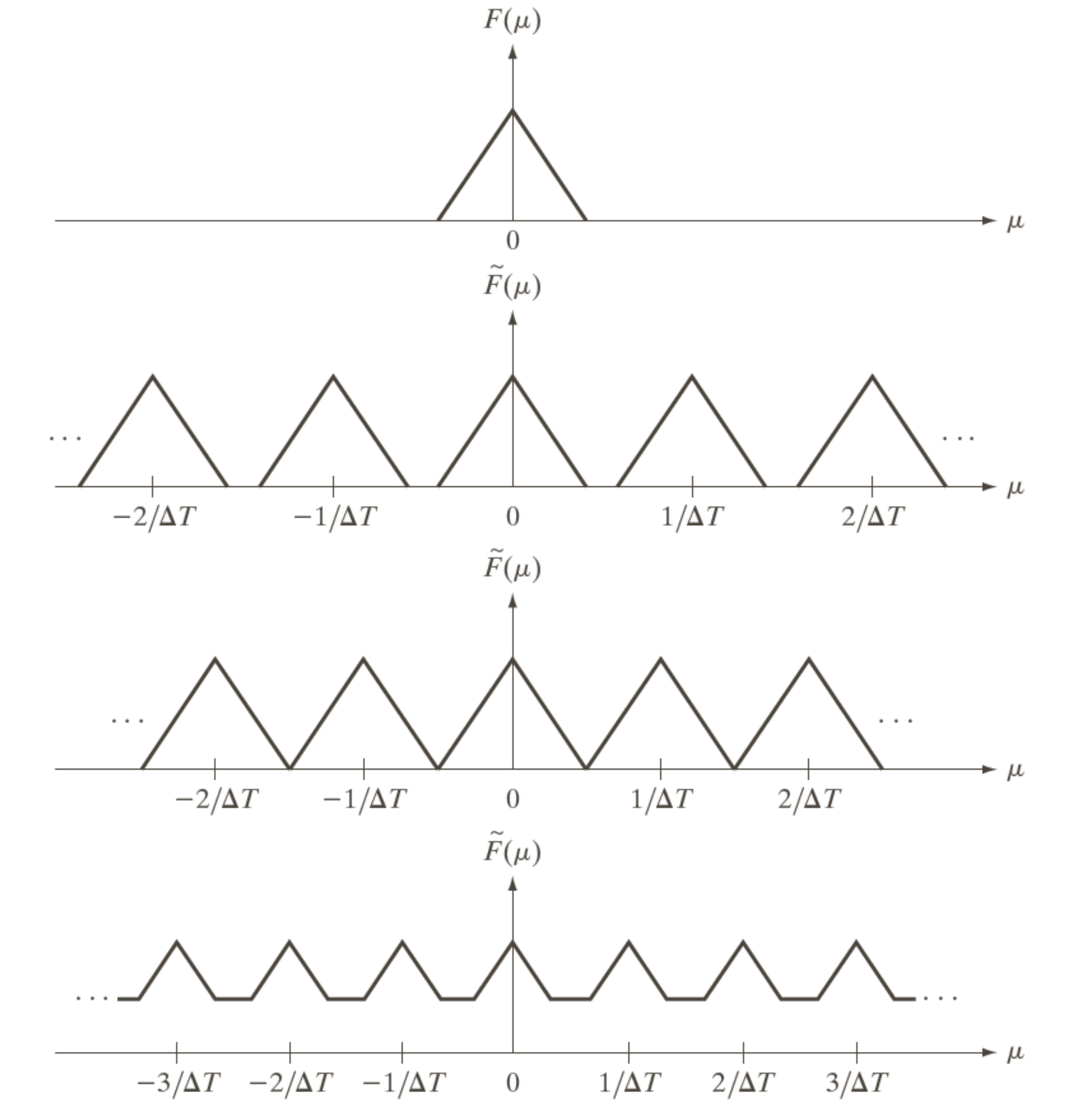


Fourier Transform of a band-limited 1D function

Sampling Theorem:

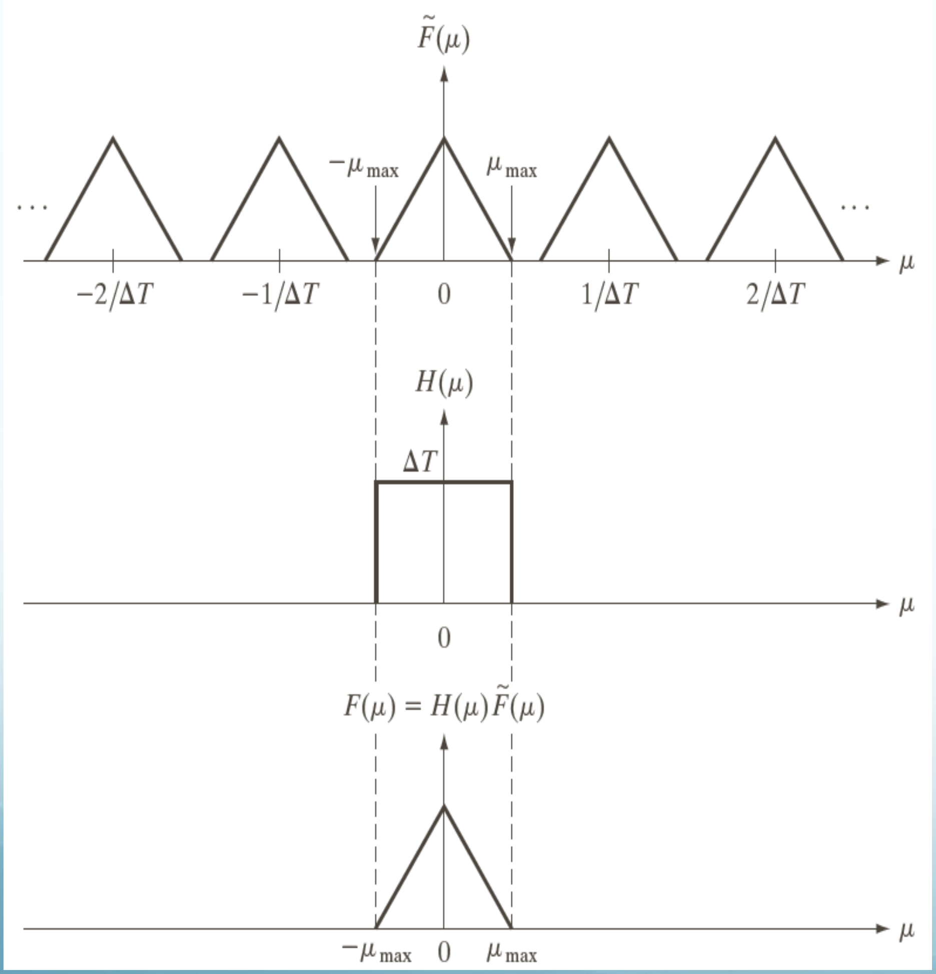
A band-limited signal should be sampled at least twice the highest frequency content of the signal, so that the signal can be RECOVERED UNIQUELY from its samples

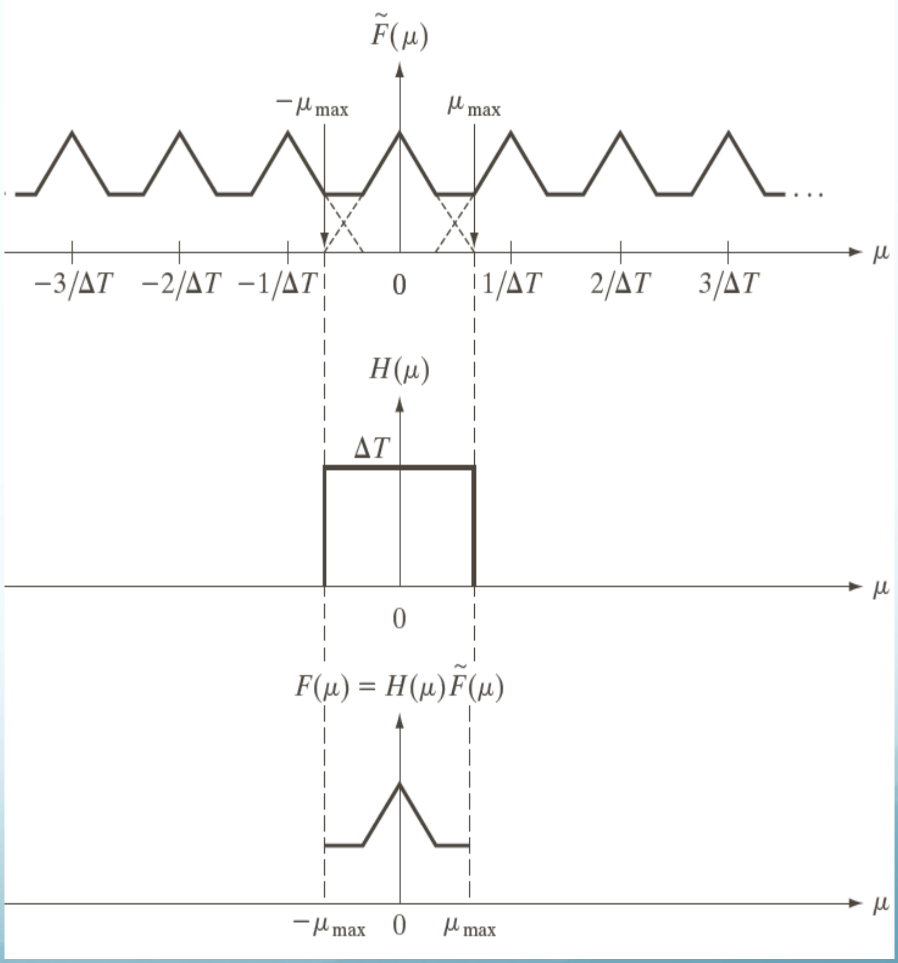
Nyquist Rate



(a) Fourier transform of a band-limited function. (b)–(d) Transforms of the corresponding sampled function under the conditions of over-sampling, critically-sampling, and under-sampling, respectively.

Reconstruction – 1D example

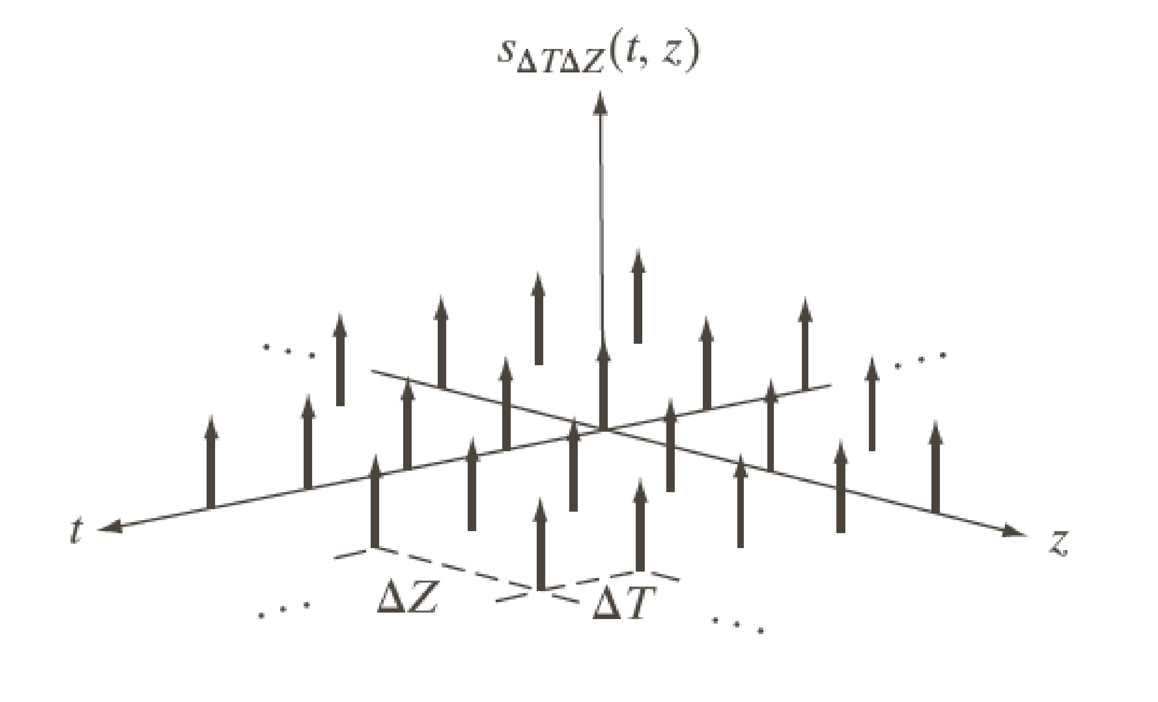




Sampling in 2D

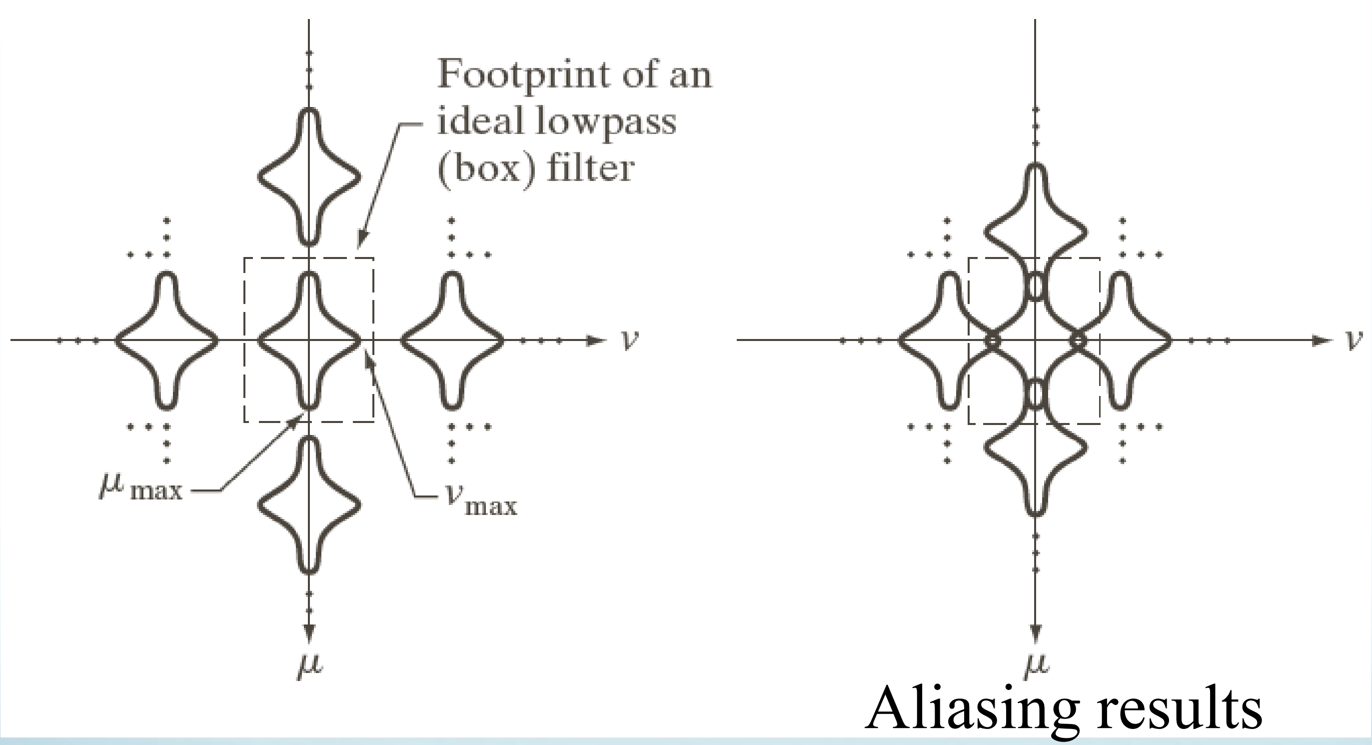
Sampling Theorem:

A band-limited signal should be sampled at least twice the highest frequency content of the signal, so that the signal can be RECOVERED UNIQUELY from its samples



Nyquist Rate should be satified in both directions

Fourier Transform of a band-limited 2D function



Two-dimensional Fourier transforms of (a) an over-sampled, and (b) under-sampled band-limited function.

Aliasing Examples

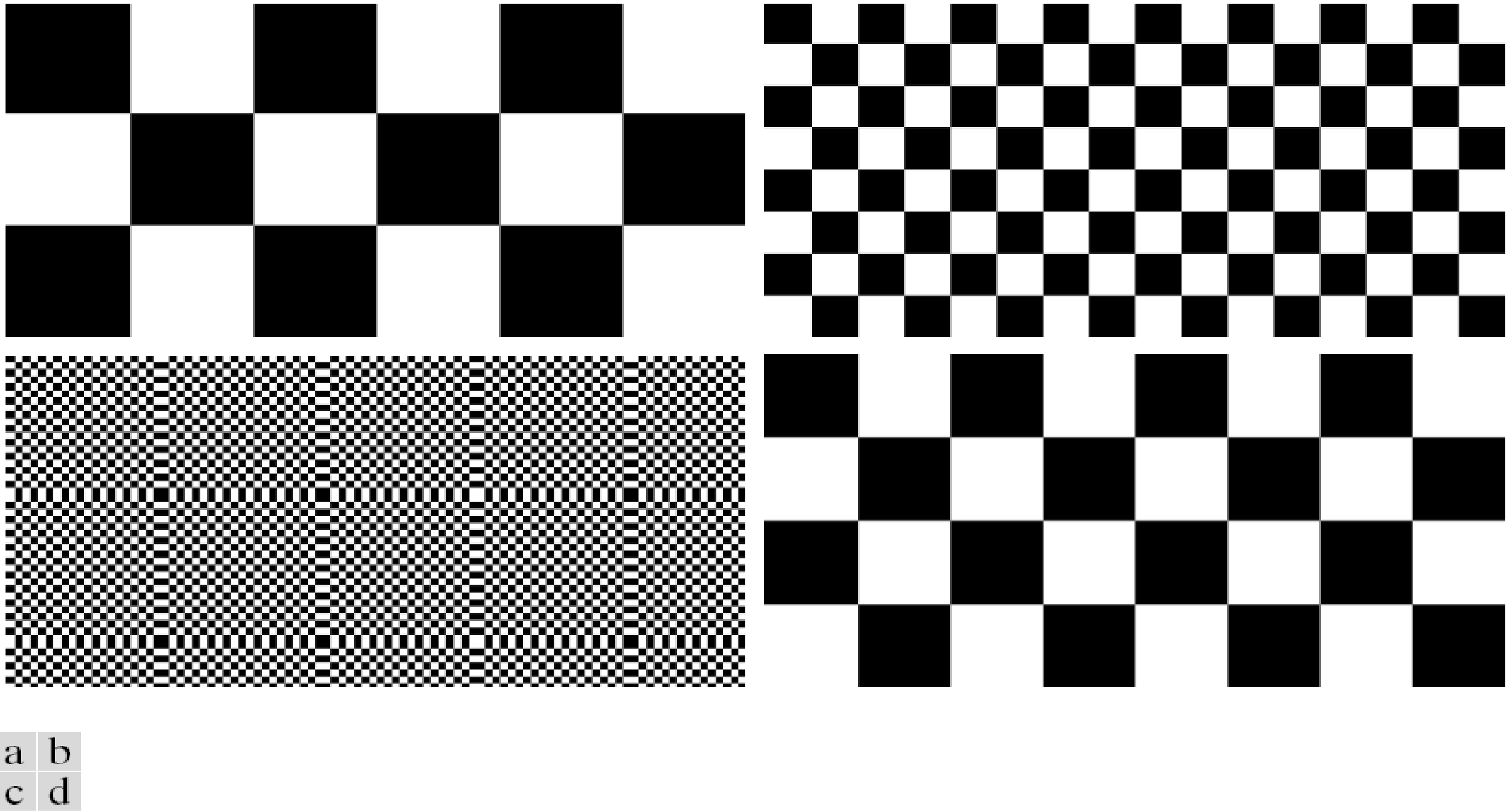


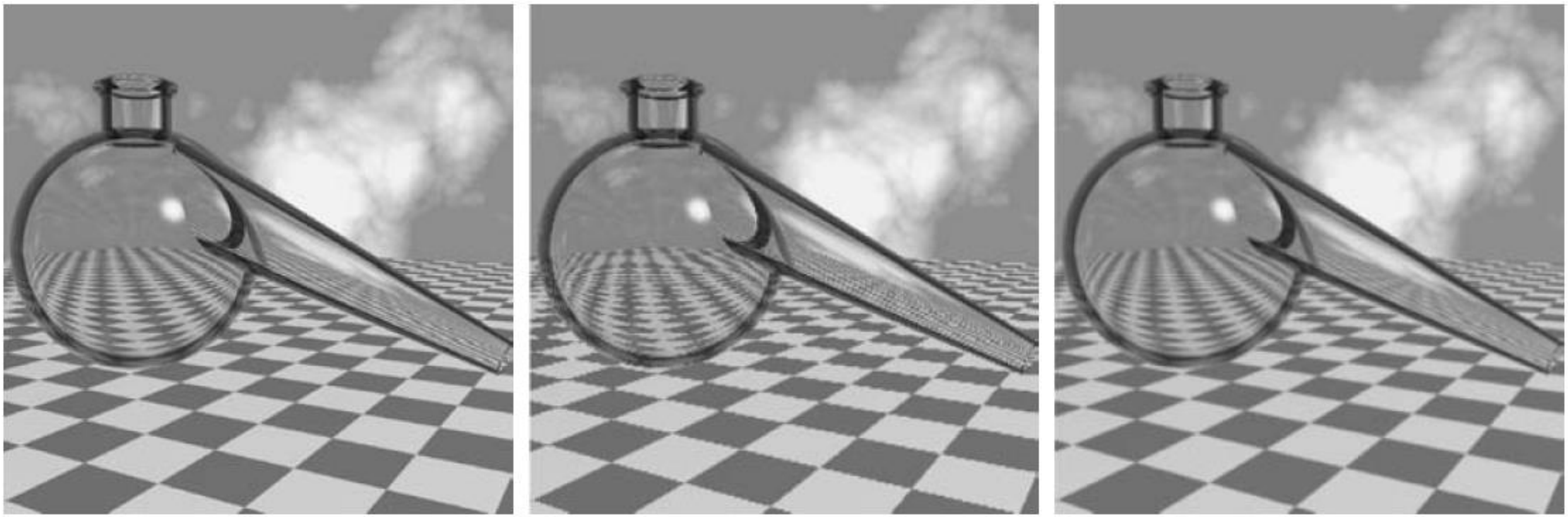
FIGURE 4.16 Aliasing in images. In (a) and (b), the lengths of the sides of the squares are 16 and 6 pixels, respectively, and aliasing is visually negligible. In (c) and (d), the sides of the squares are 0.9174 and 0.4798 pixels, respectively, and the results show significant aliasing. Note that (d) masquerades as a "normal" image.

Aliasing – Image Resampling
Shrinking = Under-sampling

a b c

FIGURE 4.17 Illustration of aliasing on resampled images. (a) A digital image with negligible visual aliasing. (b) Result of resizing the image to 50% of its original size by pixel deletion. Aliasing is clearly visible. (c) Result of blurring the image in (a) with a 3 × 3 averaging filter prior to resizing. The image is slightly more blurred than (b), but aliasing is not longer objectionable. (Original image courtesy of the Signal Compression Laboratory, University of California, Santa Barbara.)

Aliasing + Edges = Jaggies

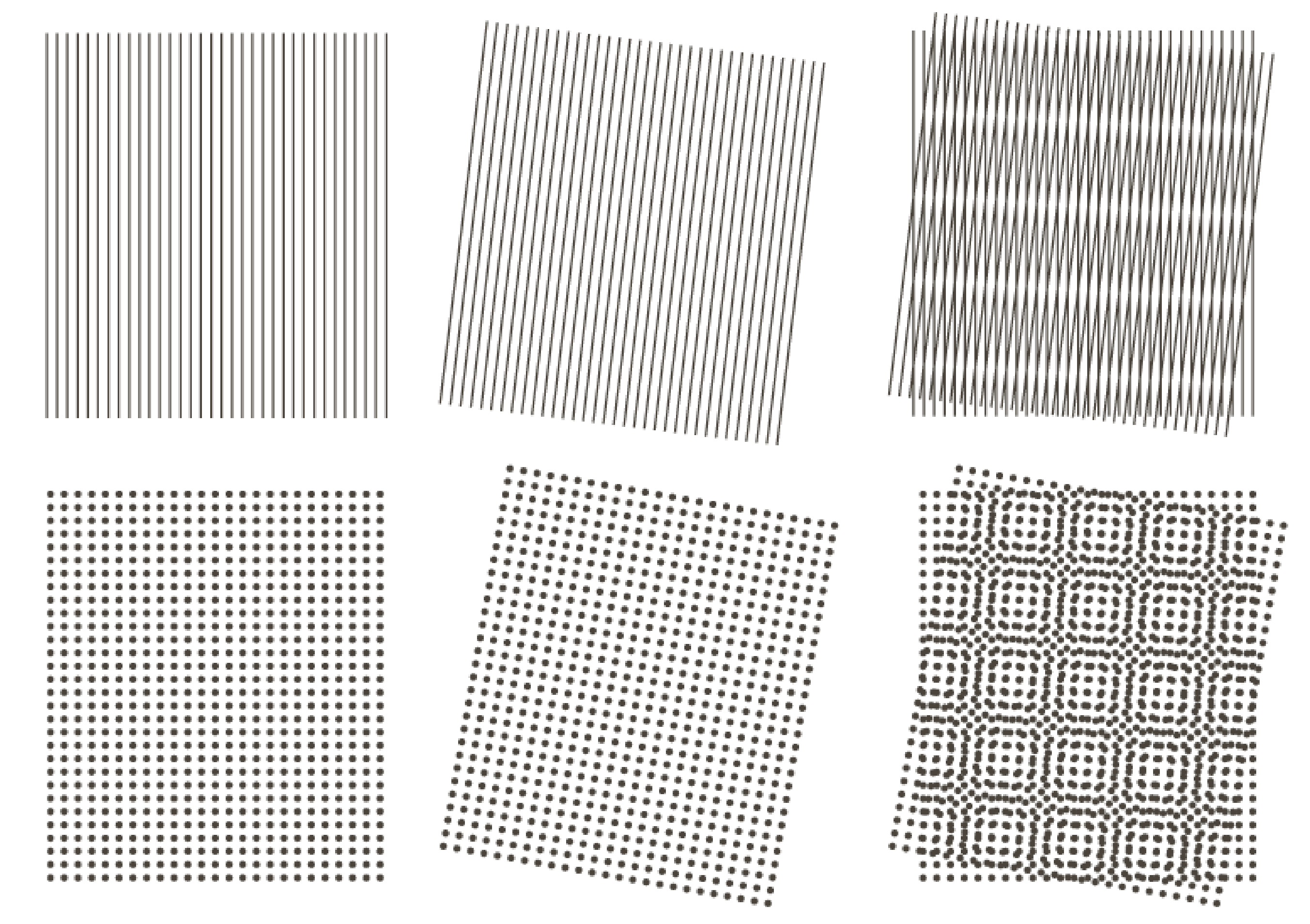


a b c

FIGURE 4.18 Illustration of jaggies. (a) A 1024×1024 digital image of a computer-generated scene with negligible visible aliasing. (b) Result of reducing (a) to 25% of its original size using bilinear interpolation. (c) Result of blurring the image in (a) with a 5×5 averaging filter prior to resizing it to 25% using bilinear interpolation. (Original image courtesy of D. P. Mitchell, Mental Landscape, LLC.)

Moiré patterns

Sampling (near) periodic components



Examples of the moiré effect. These are ink drawings, not digitized patterns. Superimposing one pattern on the other is equivalent mathematically to multiplying the patterns.

Moiré patterns

DIVERS PREPARING FOR WORK.

The fine lines that make up the sky in this image create moiré patterns when shown at some resolutions for the same reason that photographs of televisions exhibit moiré patterns: the lines are not absolutely level.

LINK

A moving moiré pattern on a billboard. The wind moves the printed net, thus constantly changing its distance to the wall and to the net's shadow.

Jean Baptiste Joseph Fourier

Fourier was born in Auxerre, France, 1768

- Most famous for his work

“*La Théorie Analitique de la Chaleur*” published in 1822

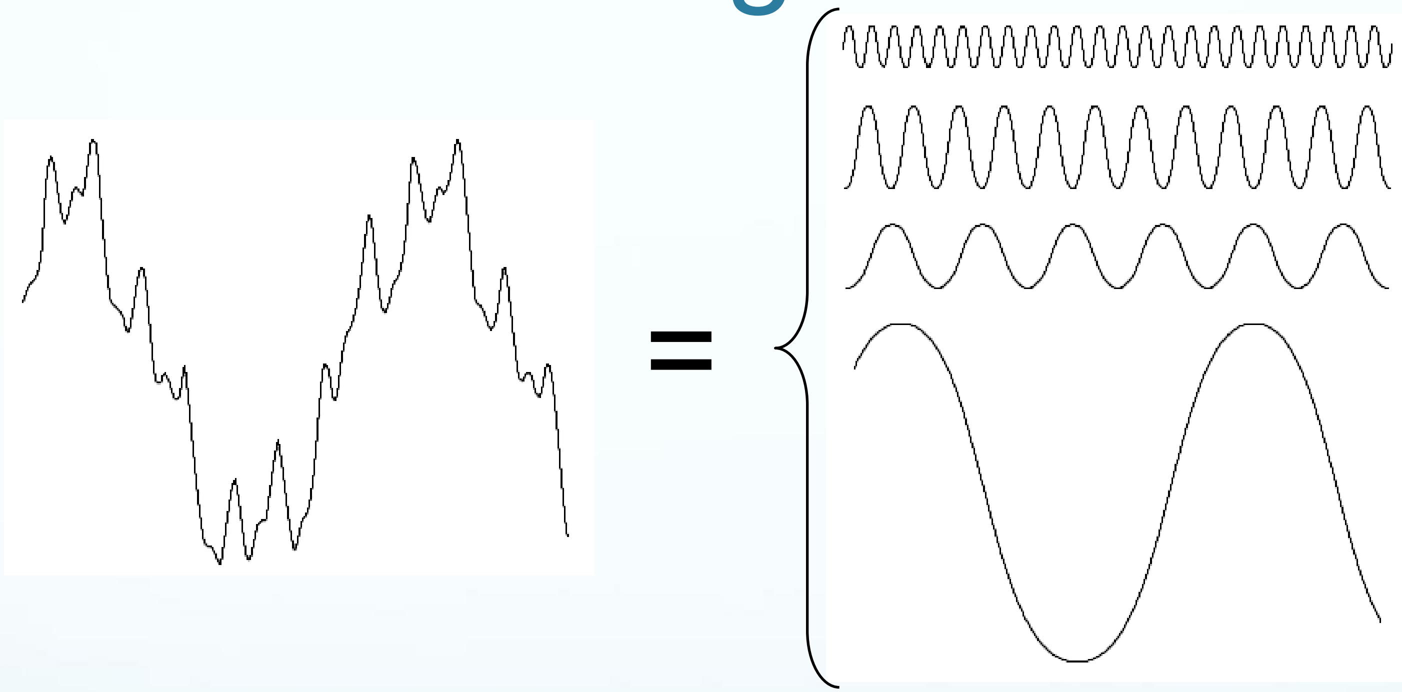
- Translated into English in 1878

“*The Analytic Theory of Heat*”

Nobody paid much attention when the work was first published

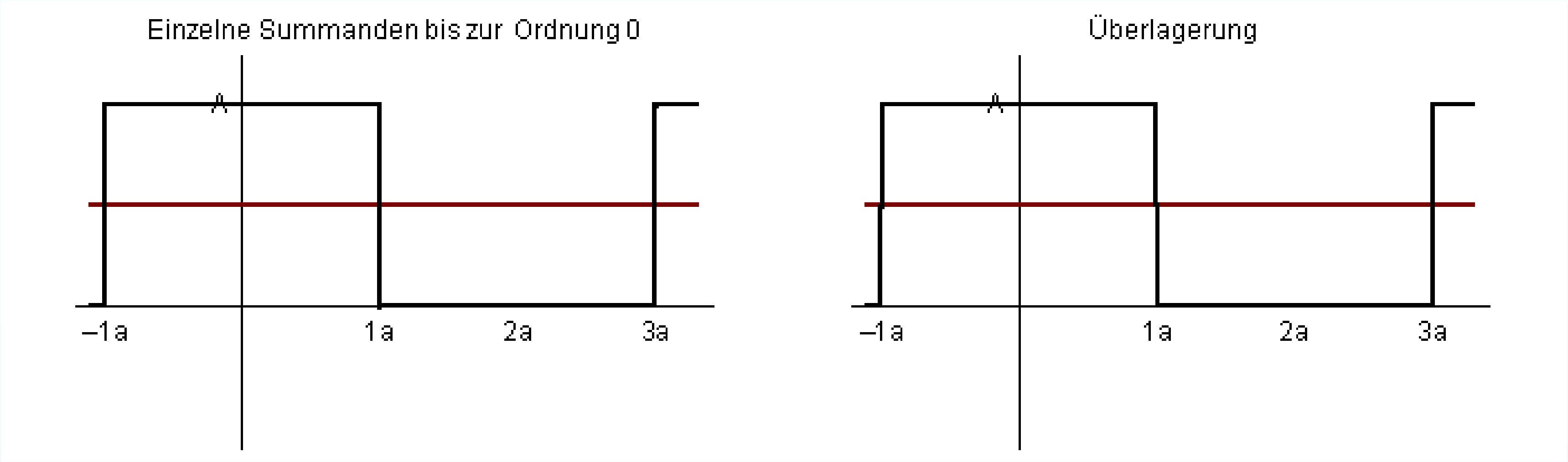
One of the most important mathematical theories in modern engineering

The Big Idea



Any function that periodically repeats itself can be expressed as a sum of sines and cosines of different frequencies each multiplied by a different coefficient – a *Fourier series*

The Big Idea



Notice how it gets closer and closer to the original function with adding more and more frequencies

Discrete Fourier Transform (DFT)

The *Discrete Fourier Transform* of $f(x, y)$, for x = 0, 1, 2…M-1 and y = 0,1,2…N-1, denoted by $F(u, v)$, is given by the equation:

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(ux/M + vy/N)}$$

for u = 0, 1, 2…M-1 and v = 0, 1, 2…N-1.

delta u = 1/(M delta_T);   delta v = 1/(N delta_Z);

Discrete Fourier Transform (DFT)

DFT and its inverse is periodic in u and v directions

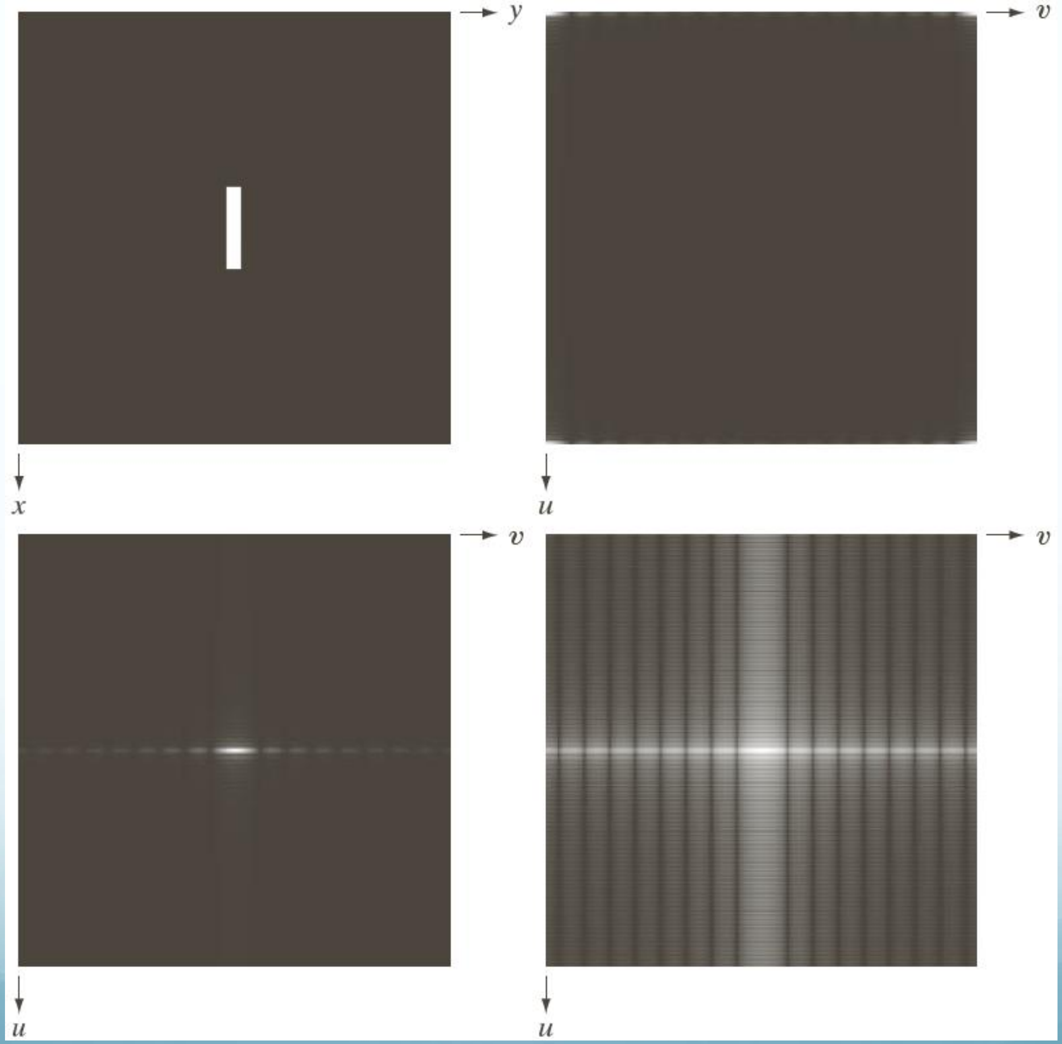
$$F(u,v)=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$$

for u = 0, 1, 2…M-1 and v = 0, 1, 2…N-1.

delta u = 1/(M delta_T); delta v = 1/(N delta_Z);

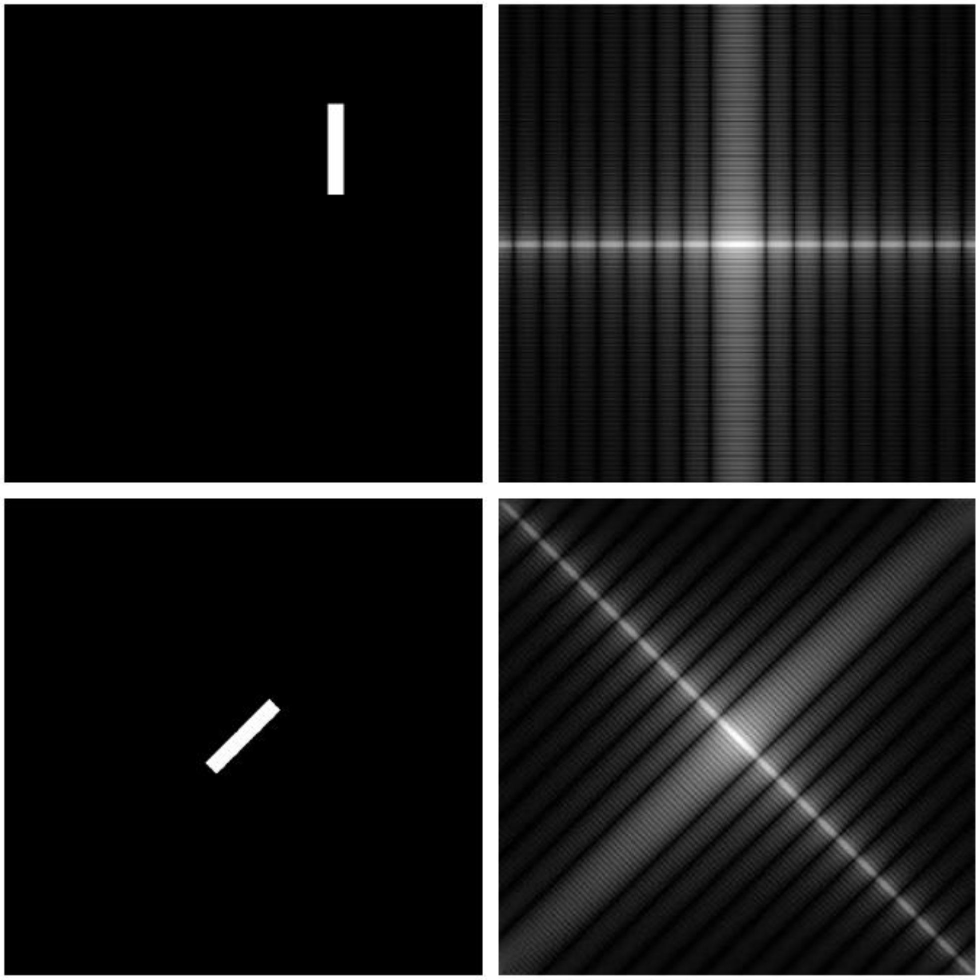
DFT & Images

The DFT of a two dimensional image can be visualised by showing the spectrum of the images component frequencies



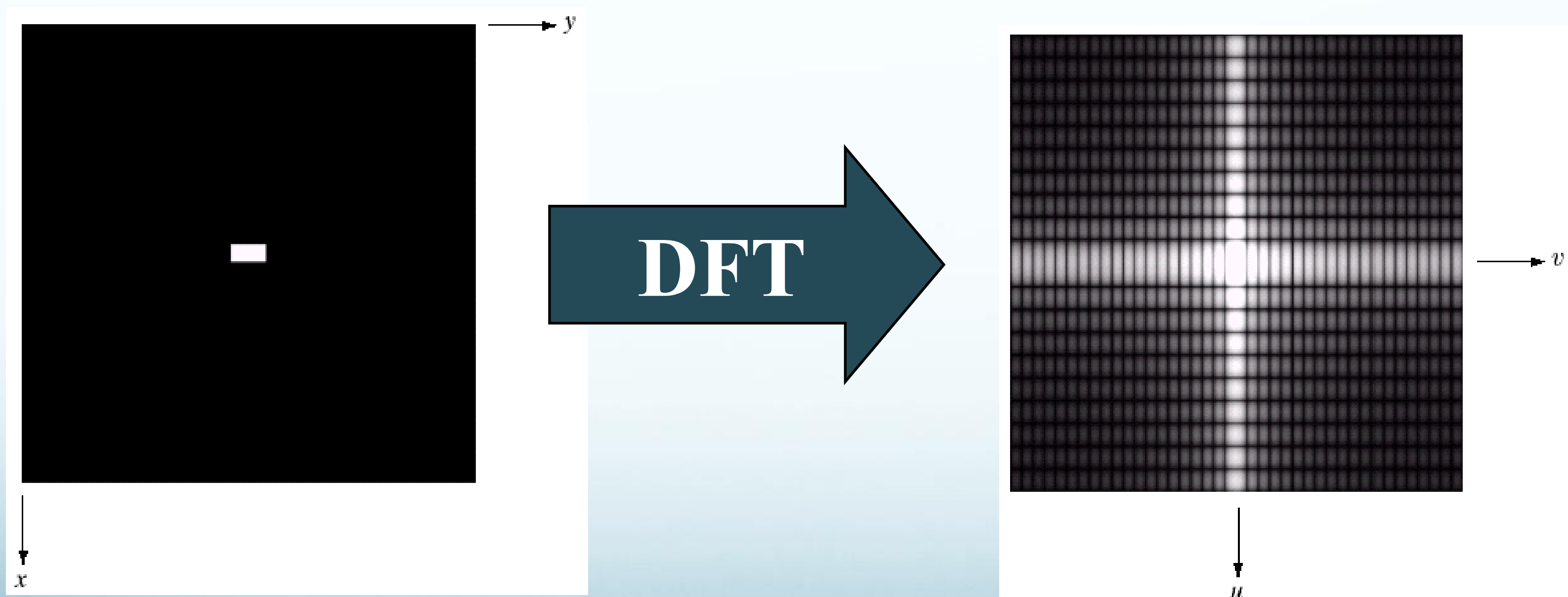
(a) Image.
(b) Spectrum showing bright spots in the four corners.
(c) Centered spectrum. (d) Result showing increased detail after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The coordinate convention used throughout the book places the origin of the spatial and frequency domains at the top left.

DFT & Images

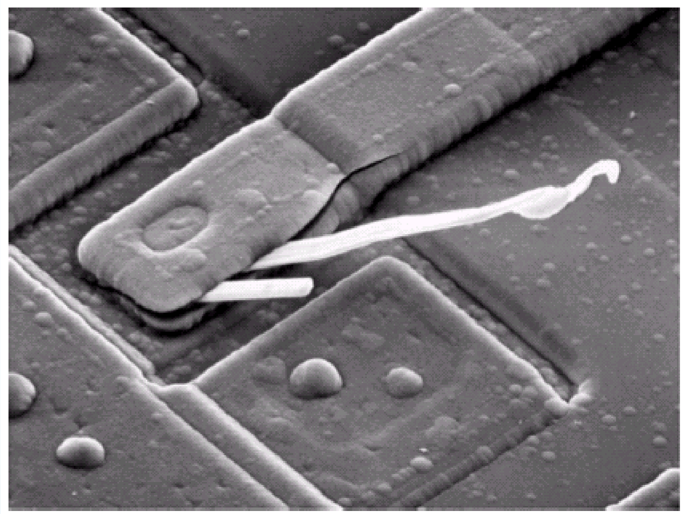


(a) The rectangle in Fig. 4.24(a) translated, and (b) the corresponding spectrum. (c) Rotated rectangle, and (d) the corresponding spectrum. The spectrum corresponding to the translated rectangle is identical to the spectrum corresponding to the original image in Fig. 4.24(a).

DFT & Images

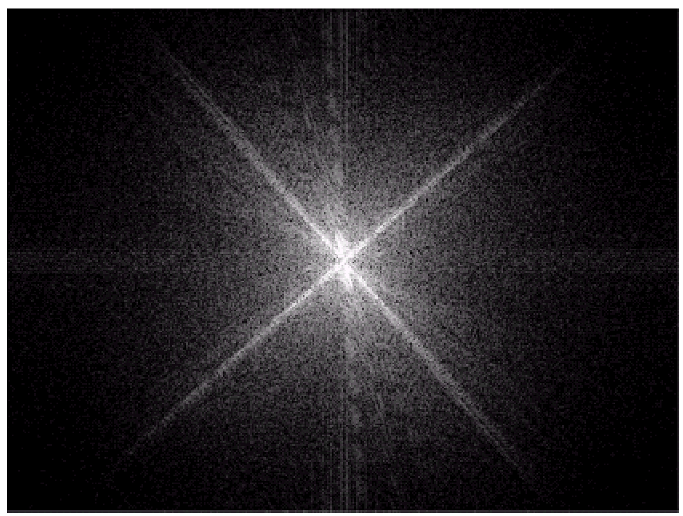


DFT & Images



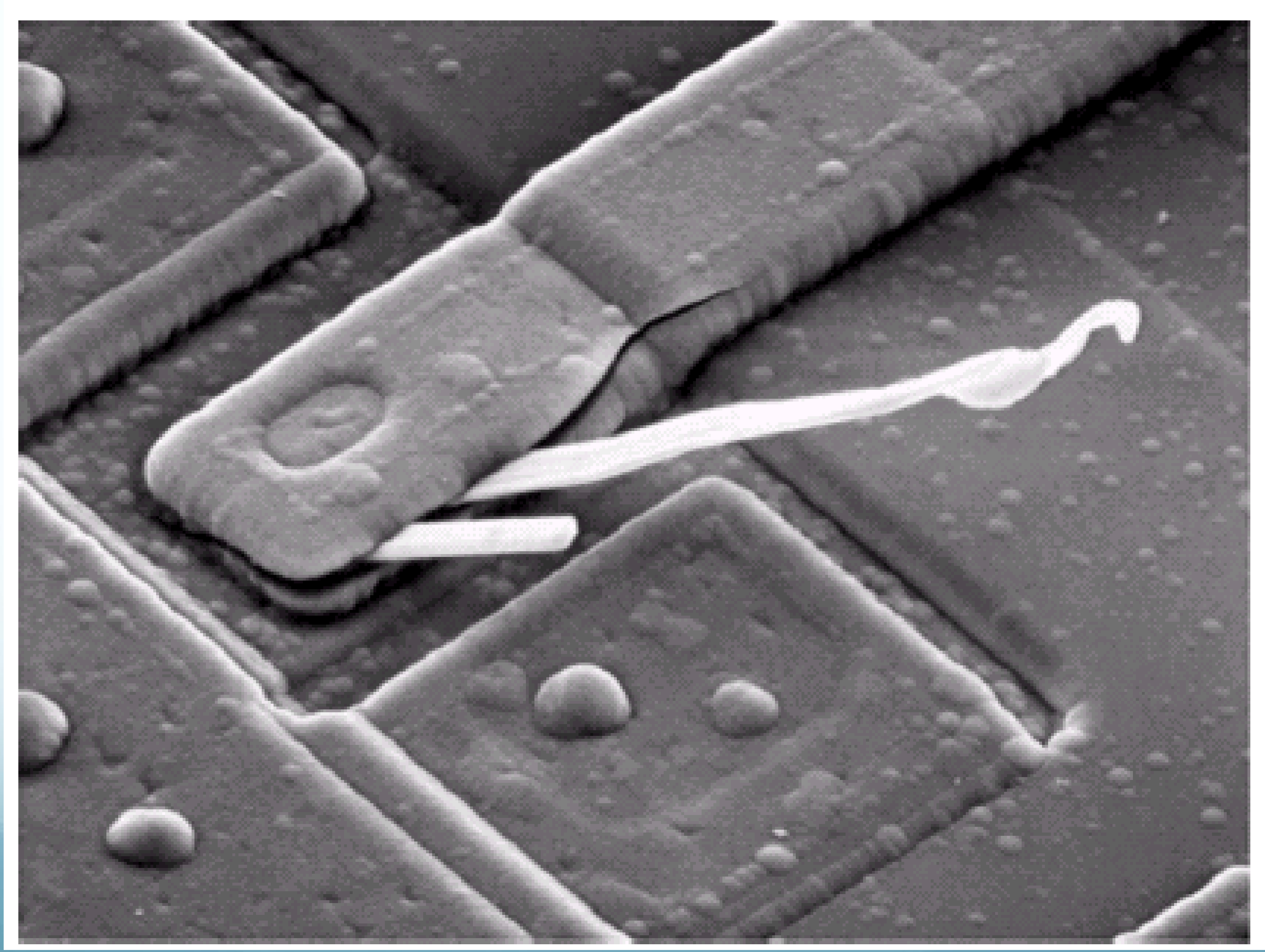
Scanning electron microscope image of an integrated circuit magnified ~2500 times

DFT

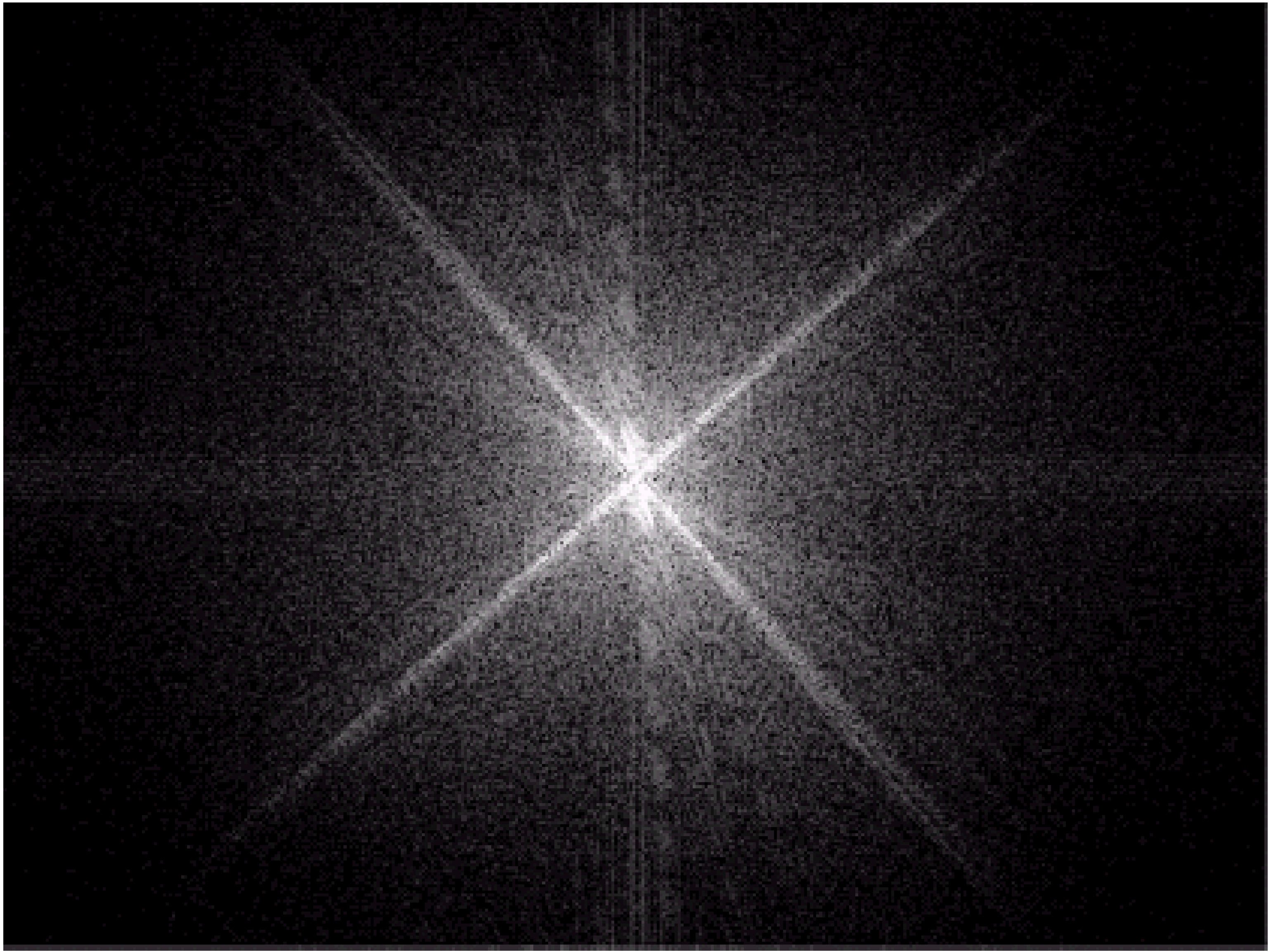


Fourier spectrum of the image

DFT & Images



DFT & Images



The Inverse DFT

It is really important to note that the Fourier transform is completely **reversible**

The inverse DFT is given by:

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi(ux/M + vy/N)}$$

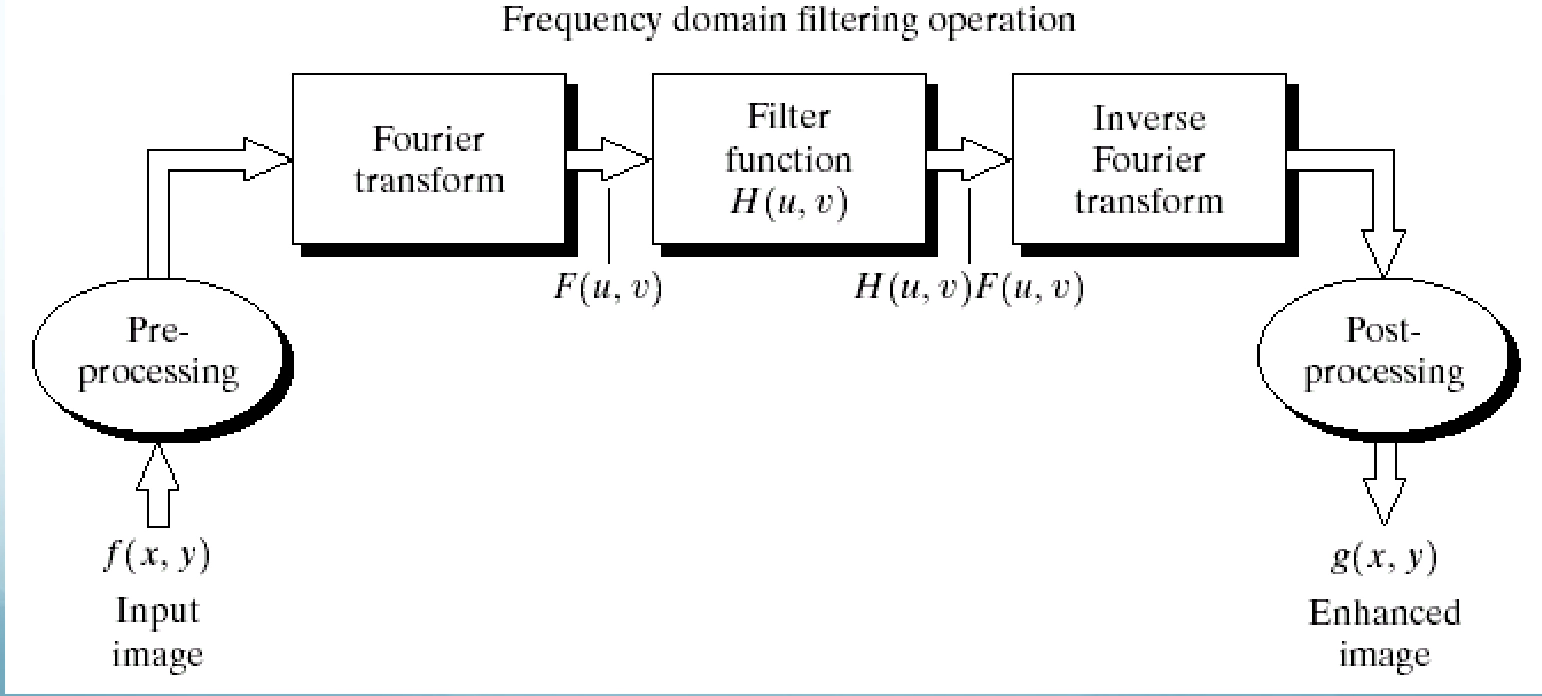
for x = 0, 1, 2…M-1 and y = 0, 1, 2…N-1

DFT and Image Processing

To filter an image in the frequency domain:

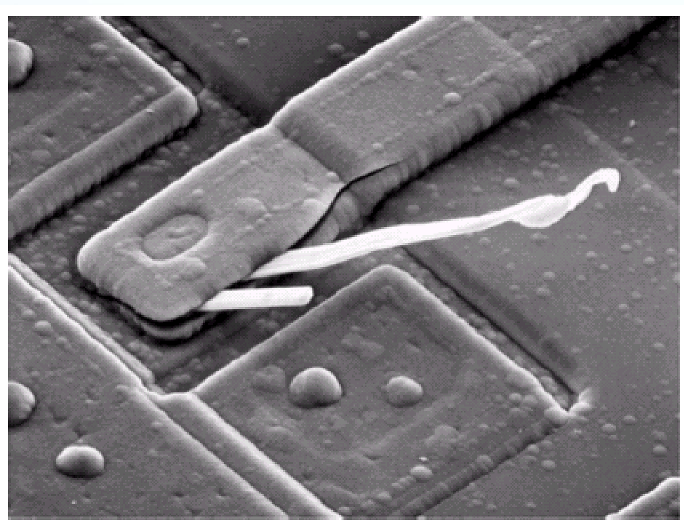
1. Compute $F(u,v)$ the DFT of the image
2. Multiply $F(u,v)$ by a filter function $H(u,v)$
3. Compute the inverse DFT of the result

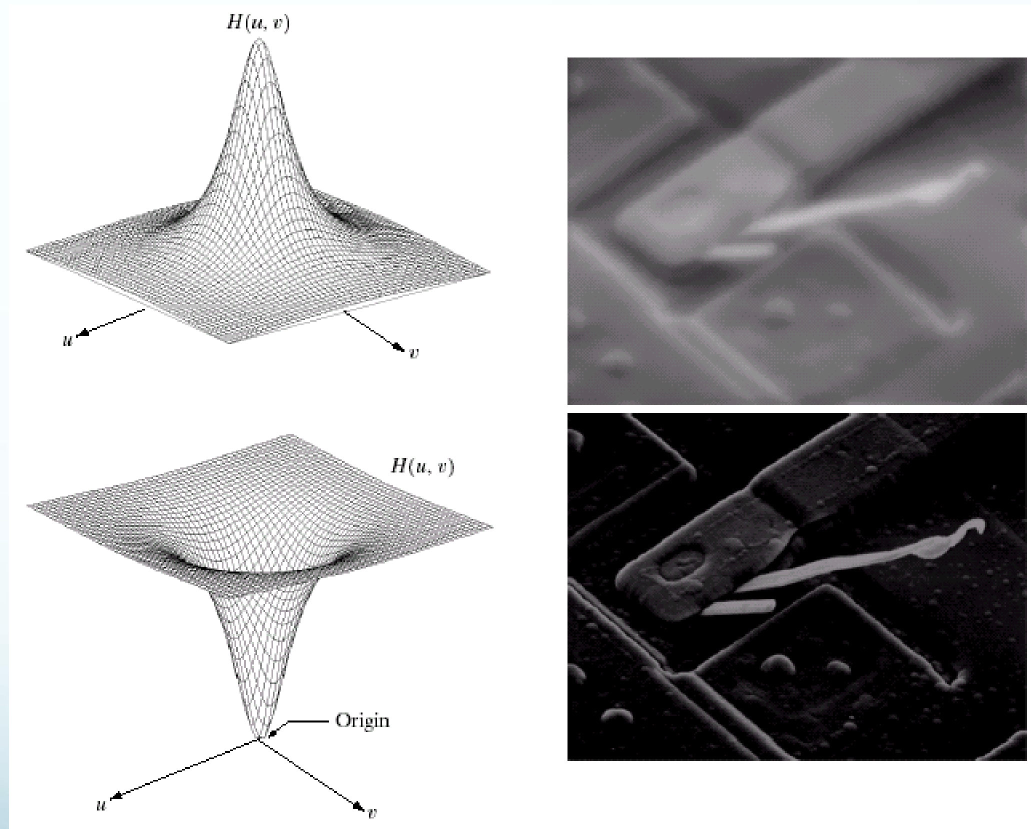
Convolution CORRESPONDS to multiplication!!!



Some Basic Frequency Domain Filters

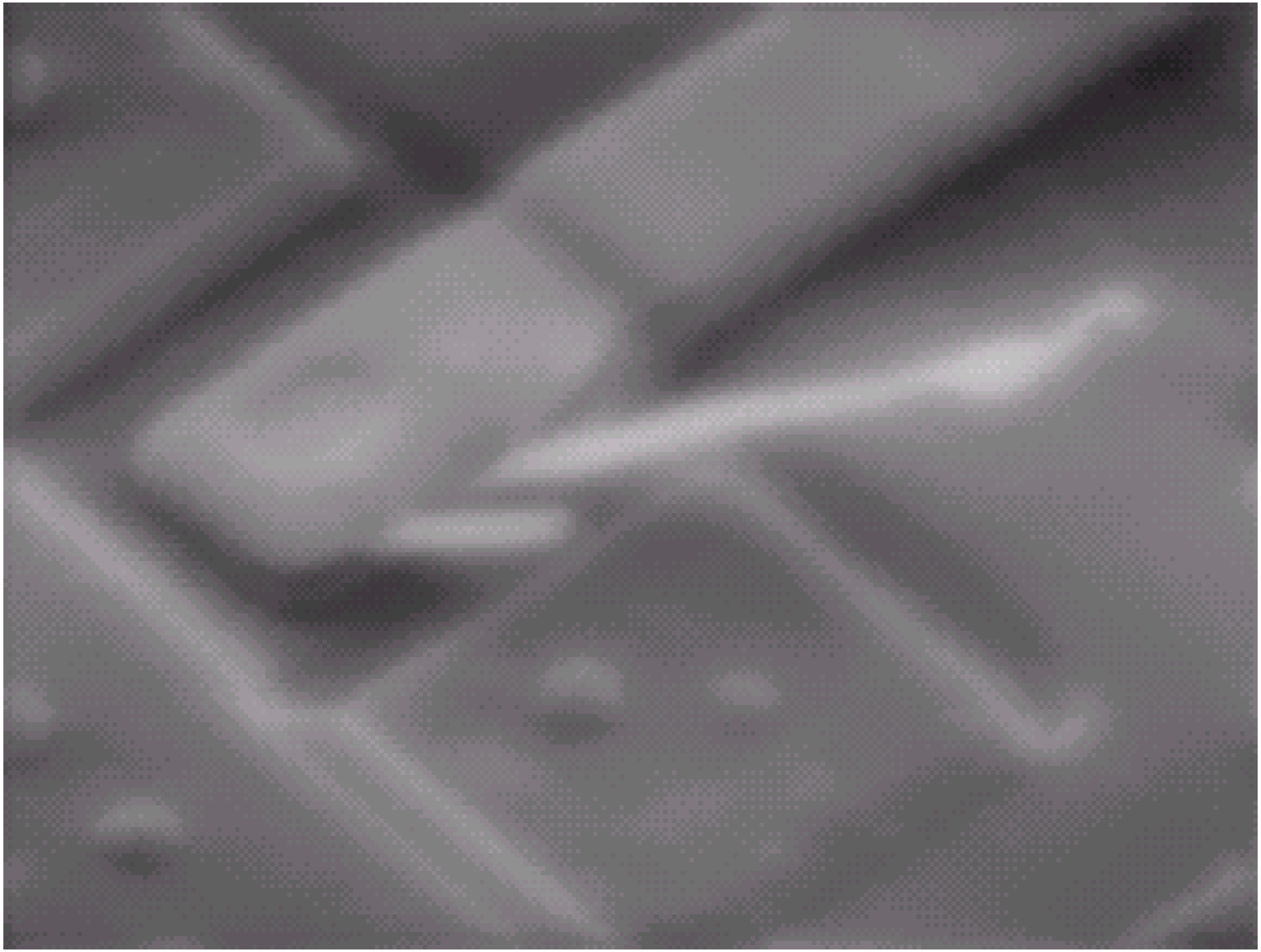
Low Pass Filter



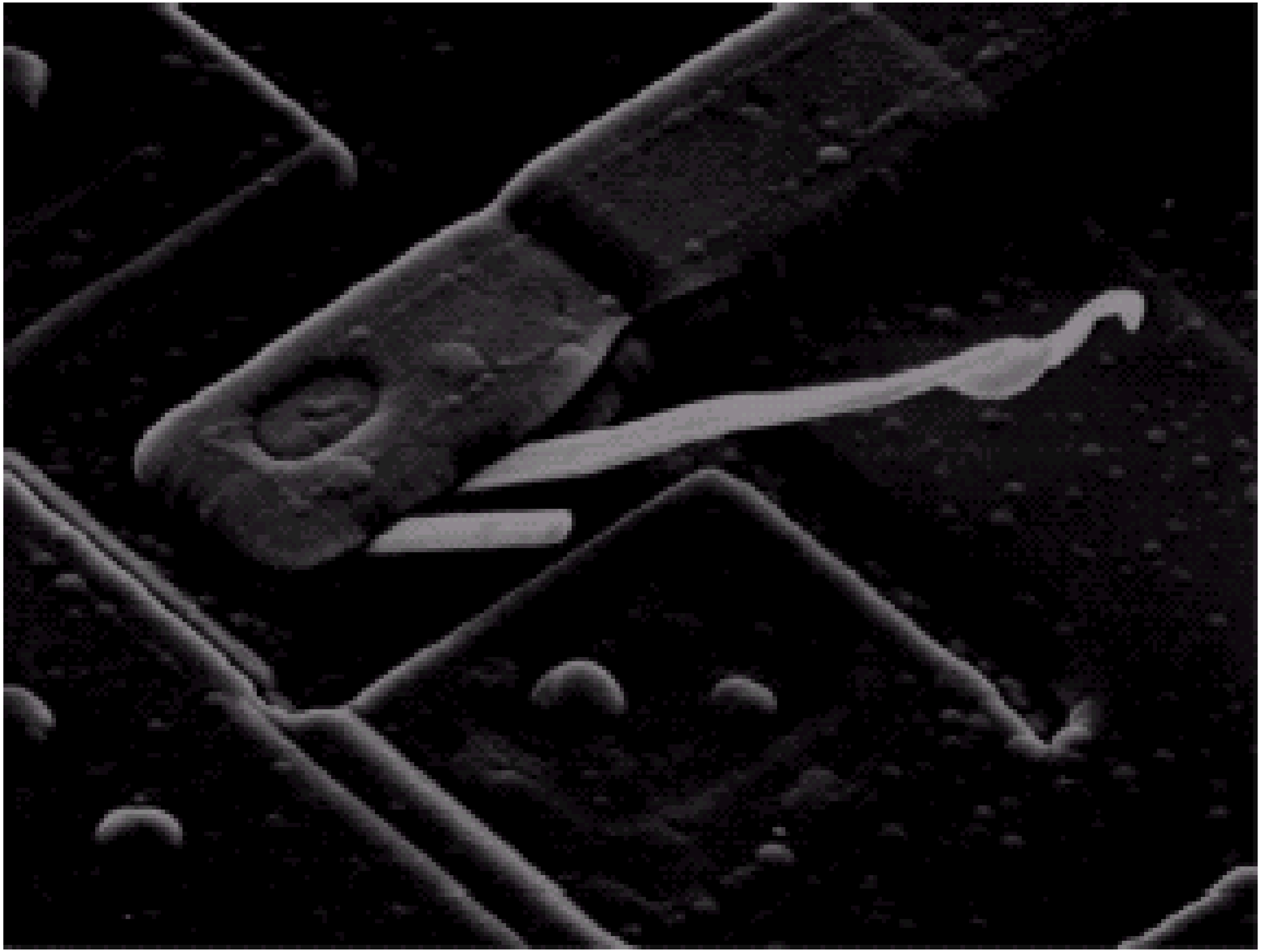


High Pass Filter

Some Basic Frequency Domain Filters



Some Basic Frequency Domain Filters



Smoothing Frequency Domain Filters

Smoothing is achieved in the frequency domain by dropping out the high frequency components

The basic model for filtering is:

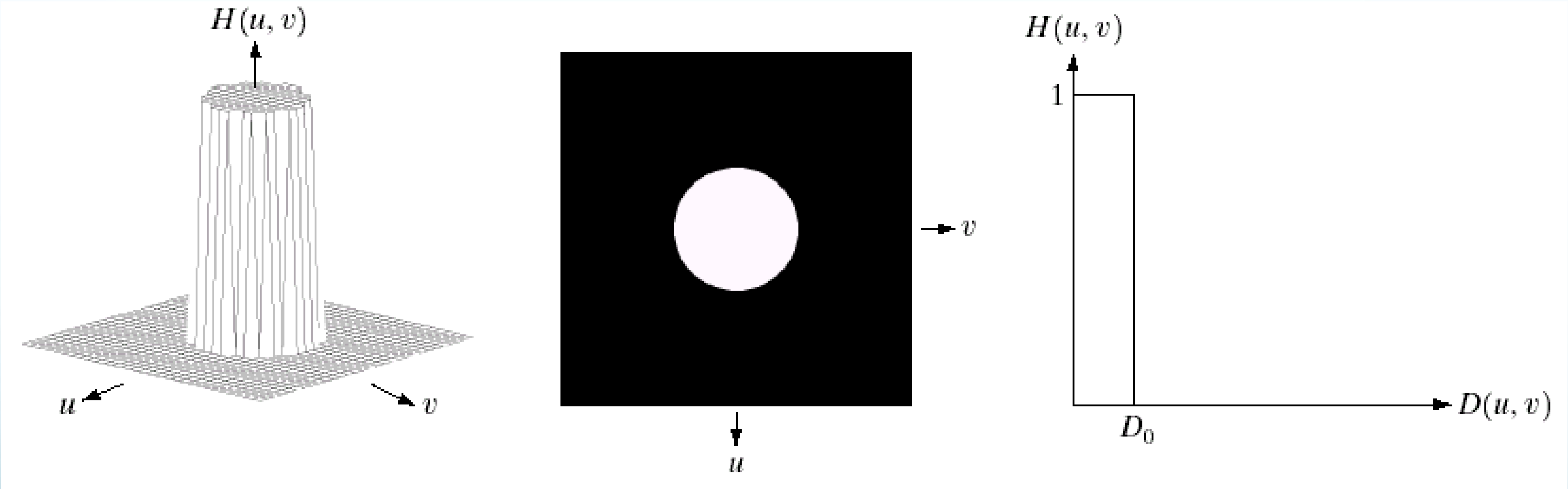
$$G(u,v) = H(u,v)F(u,v)$$

where $F(u,v)$ is the Fourier transform of the image being filtered and $H(u,v)$ is the filter transform function

Low pass filters – only pass the low frequencies, drop the high ones

Ideal Low Pass Filter

Simply cut off all high frequency components that are a specified distance D_0 from the origin of the transform



Changing the distance changes the behaviour of the filter

Ideal Low Pass Filter

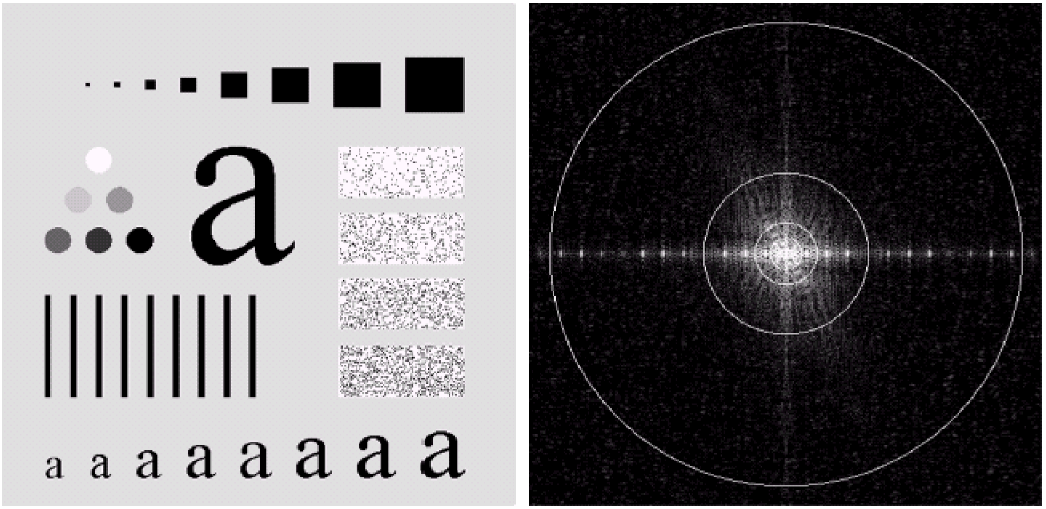
The transfer function for the ideal low pass filter can be given as:

$$H(u,v)=\begin{cases}1 & \text{if } D(u,v)\le D_0\\ 0 & \text{if } D(u,v)>D_0\end{cases}$$

where $D(u,v)$ is given as:

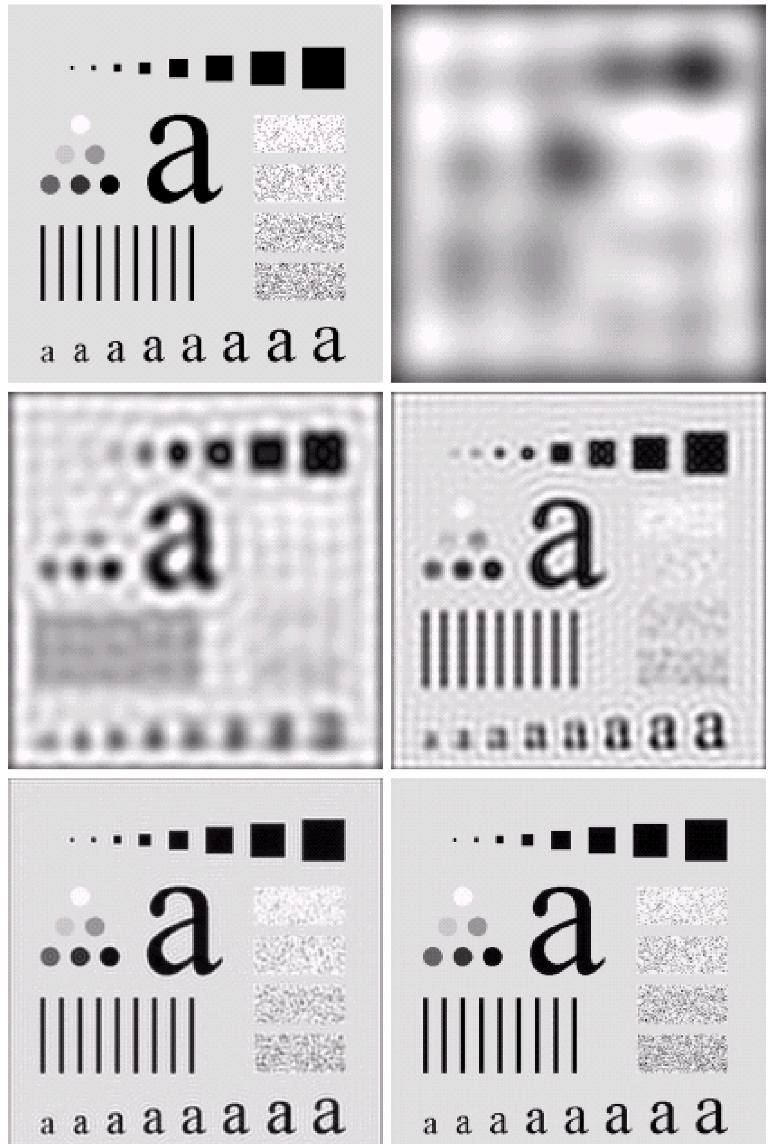
$$D(u,v)=[(u-M/2)^2+(v-N/2)^2]^{1/2}$$

Ideal Low Pass Filter



Above we show an image, it’s Fourier spectrum and a series of ideal low pass filters of radius 5, 15, 30, 80 and 230 superimposed on top of it

Ideal Low Pass Filter



Original image

Result of filtering with ideal low pass filter of radius 5

Result of filtering with ideal low pass filter of radius 15

Result of filtering with ideal low pass filter of radius 30

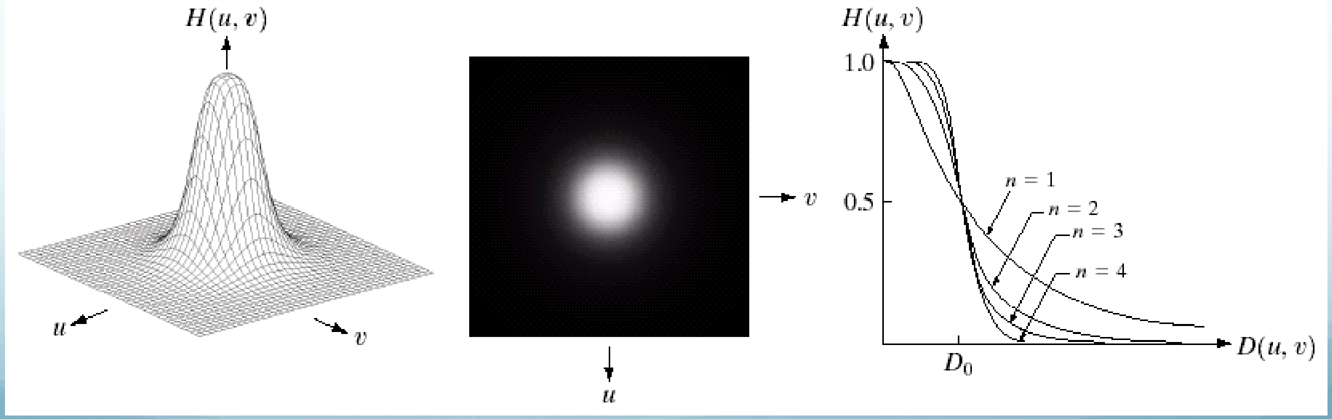
Result of filtering with ideal low pass filter of radius 80

Result of filtering with ideal low pass filter of radius 230

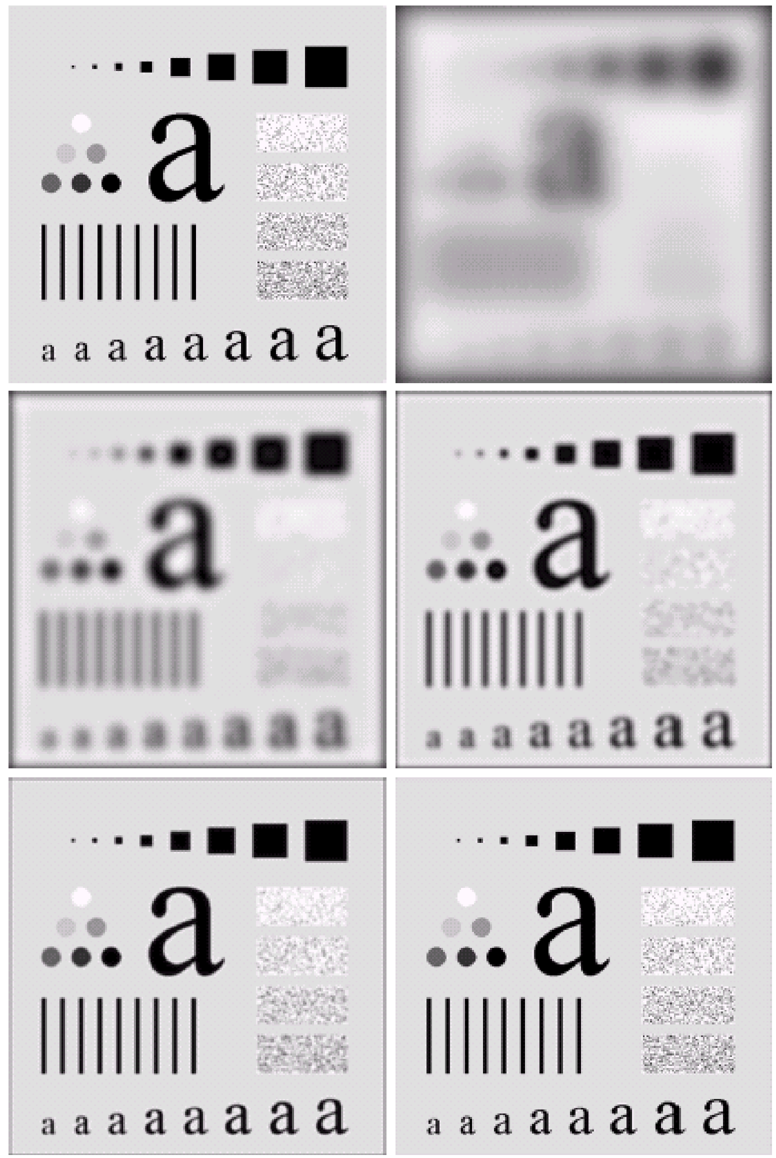
Butterworth Lowpass Filters

The transfer function of a Butterworth lowpass filter of order n with cutoff frequency at distance D_0 from the origin is defined as:

$$H(u,v) = \frac{1}{1+[D(u,v)/D_0]^{2n}}$$



Butterworth Lowpass Filter



Original image

Result of filtering with Butterworth filter of order 2 and cutoff radius 5

Result of filtering with Butterworth filter of order 2 and cutoff radius 15

Result of filtering with Butterworth filter of order 2 and cutoff radius 30

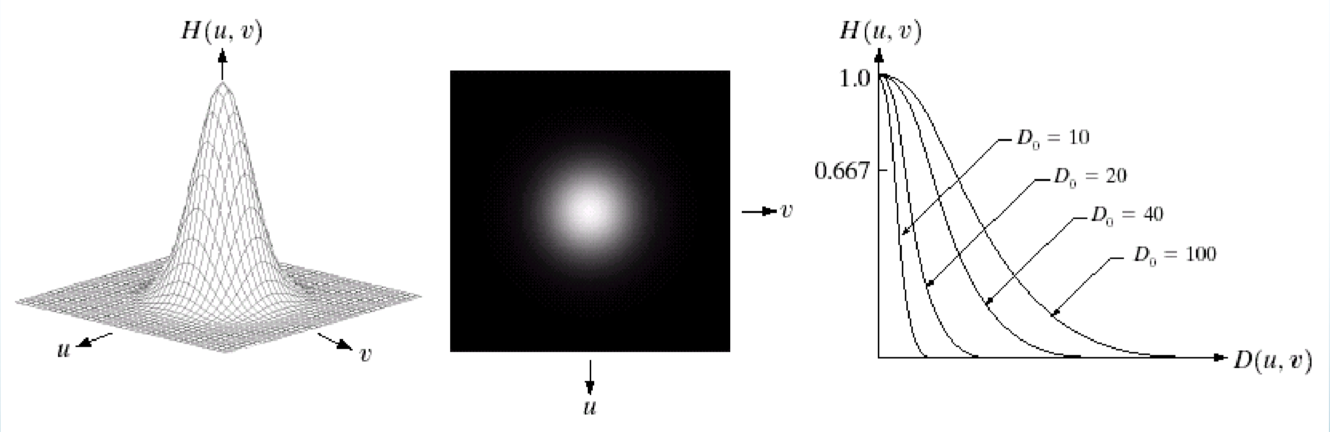
Result of filtering with Butterworth filter of order 2 and cutoff radius 80

Result of filtering with Butterworth filter of order 2 and cutoff radius 230

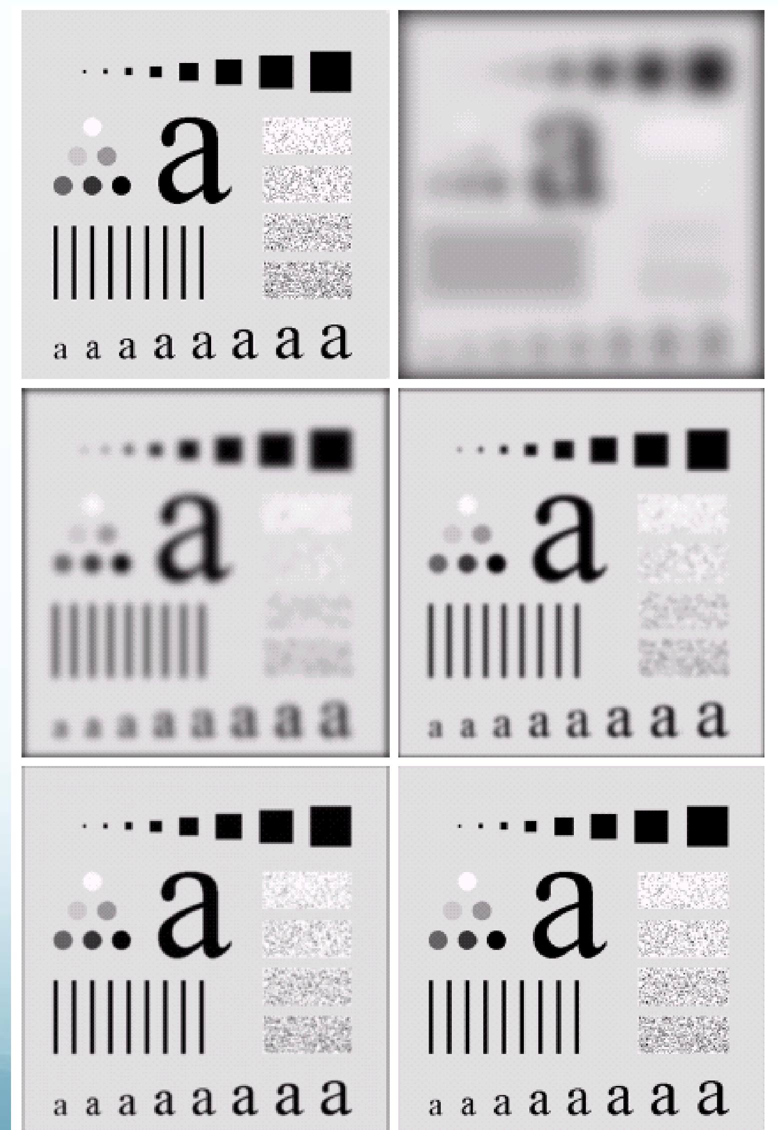
Gaussian Lowpass Filters

The transfer function of a Gaussian lowpass filter is defined as:

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$



Gaussian Lowpass Filters



Original image

Result of filtering with Gaussian filter with cutoff radius 5

Result of filtering with Gaussian filter with cutoff radius 15

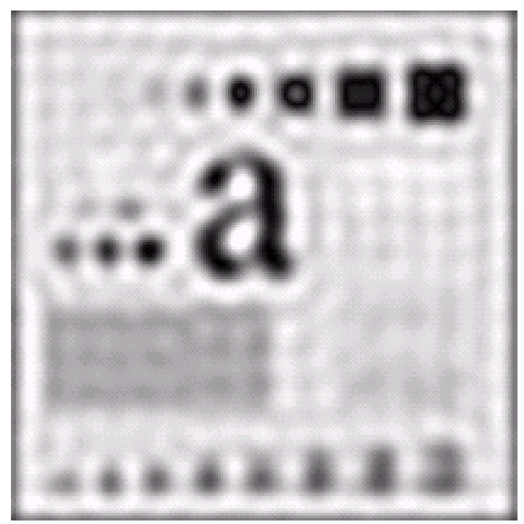
Result of filtering with Gaussian filter with cutoff radius 30

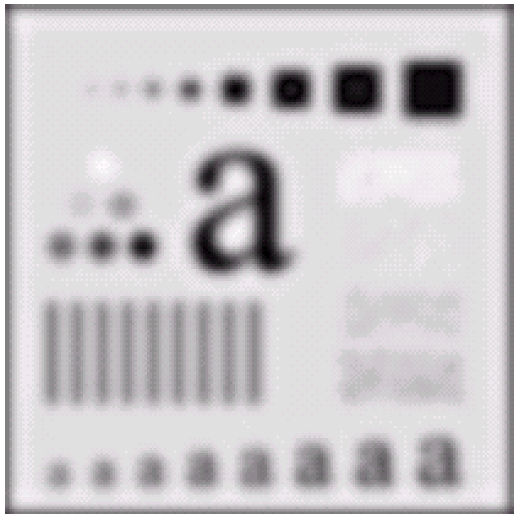
Result of filtering with Gaussian filter with cutoff radius 85

Result of filtering with Gaussian filter with cutoff radius 230

Lowpass Filters Compared

Result of filtering with ideal low pass filter of radius 15



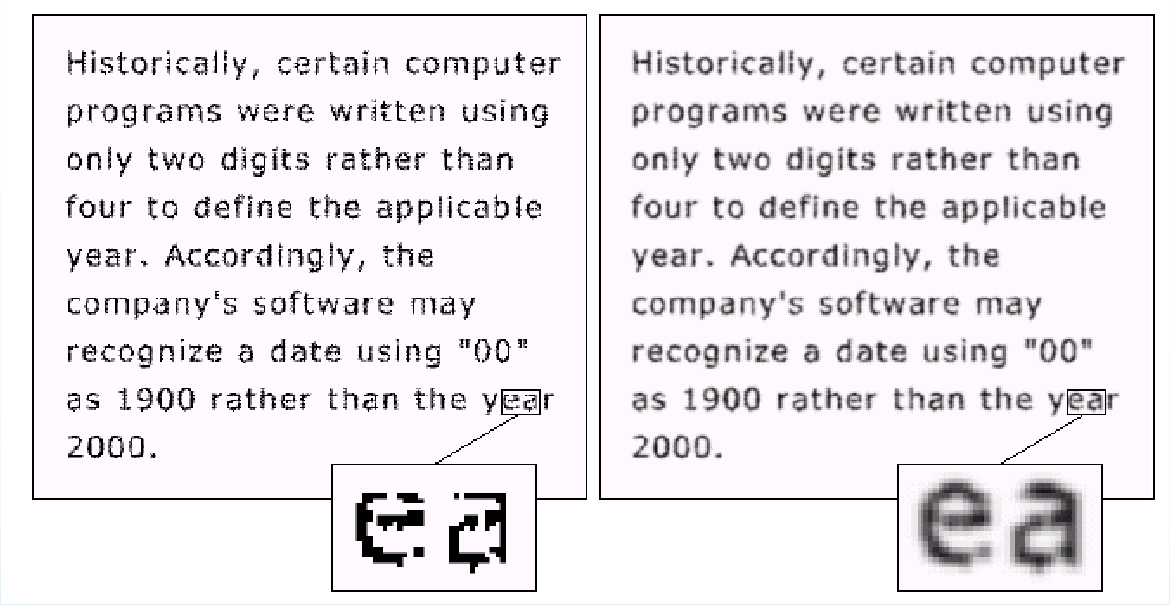


Result of filtering with Butterworth filter of order 2 and cutoff radius 15

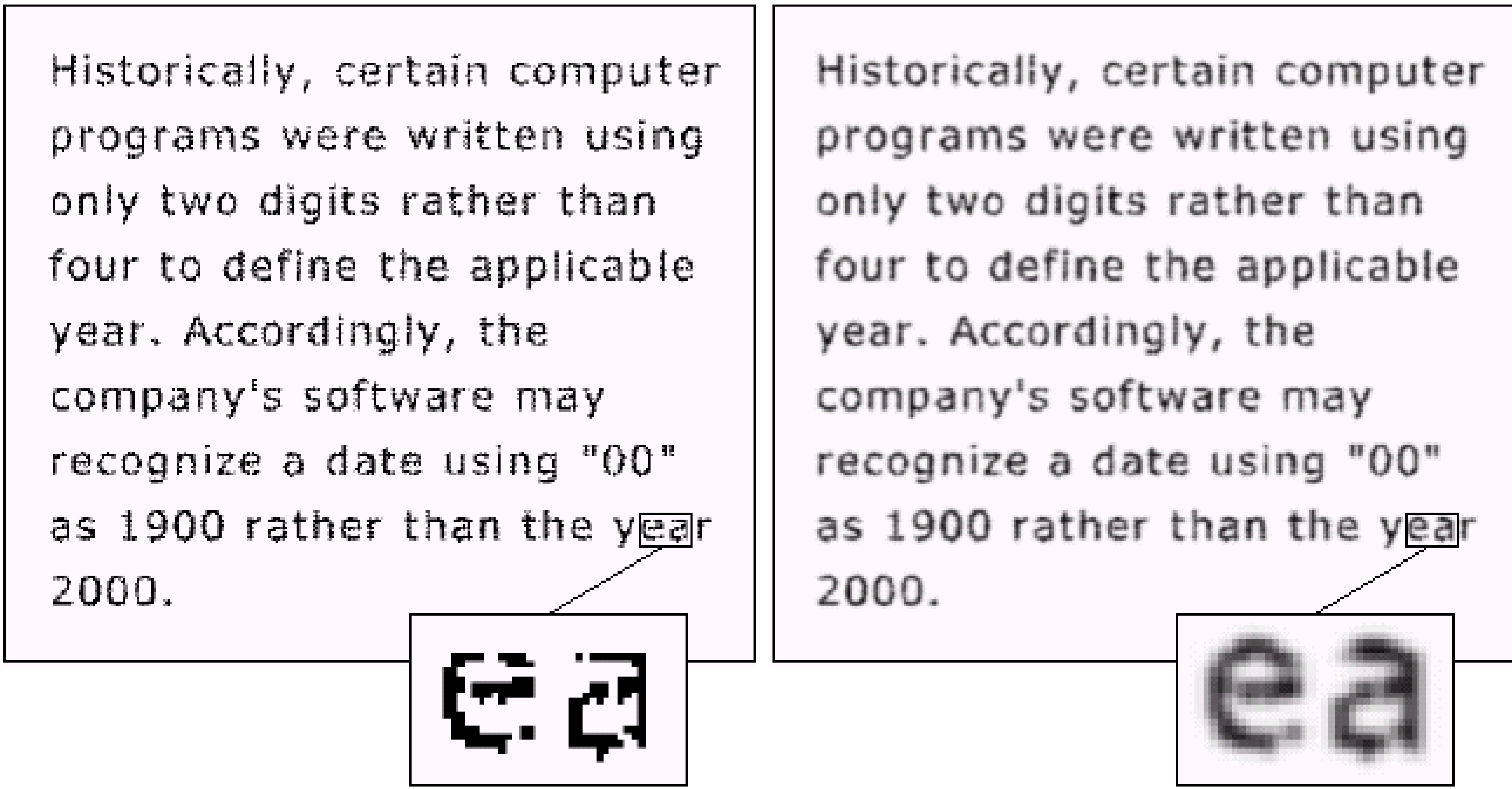
Result of filtering with Gaussian filter with cutoff radius 15

Lowpass Filtering Examples

A low pass Gaussian filter is used to connect broken text



Lowpass Filtering Examples



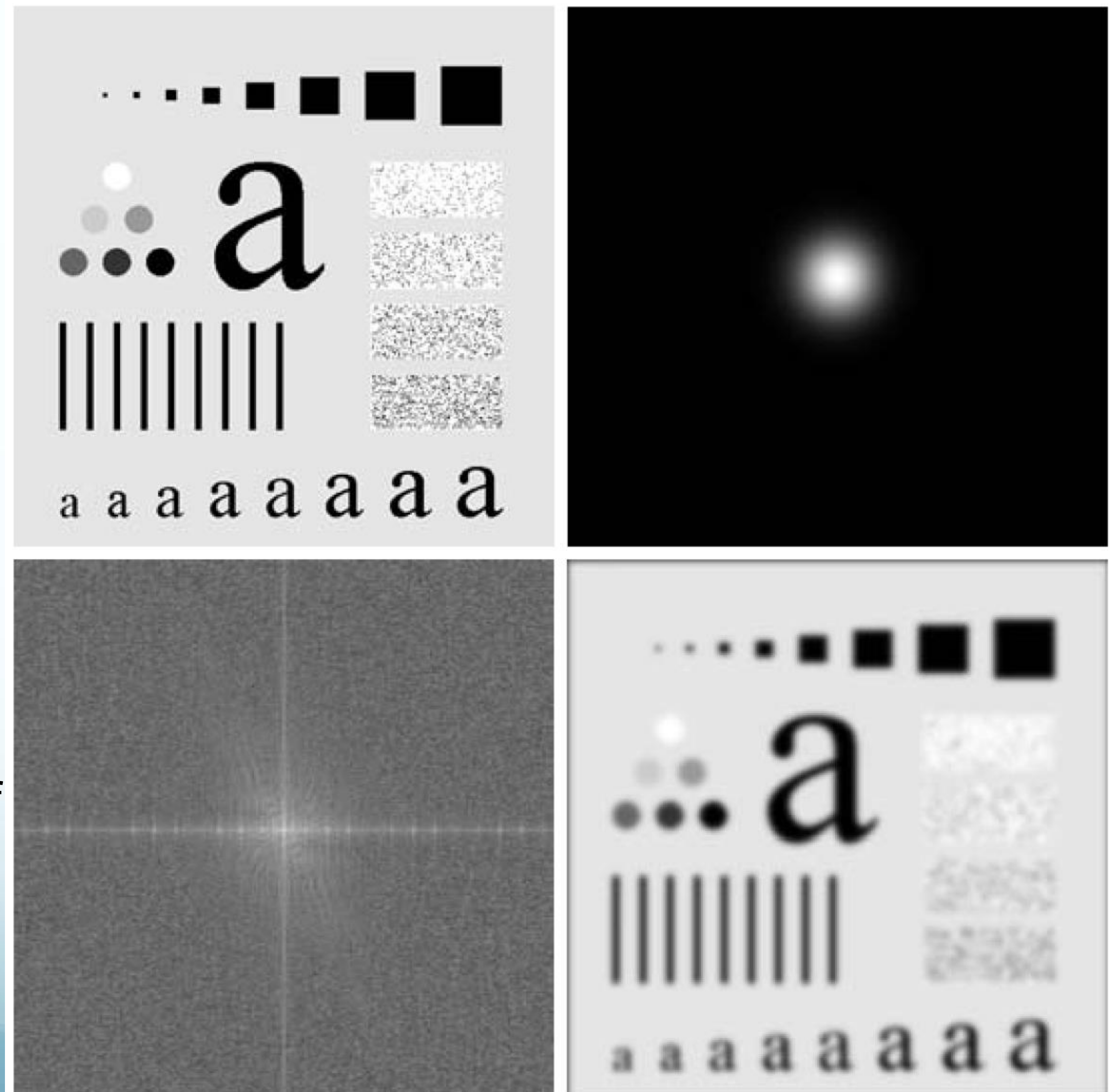
Lowpass Filtering Examples

Different lowpass Gaussian filters used to remove blemishes in a photograph

Lowpass Filtering Examples

Original image

Gaussian lowpass filter



Spectrum of original image

Processed image

Sharpening in the Frequency Domain

Edges and fine detail in images are associated with high frequency components

High pass filters – only pass the high frequencies, drop the low ones

High pass frequencies are precisely the reverse of low pass filters, so:

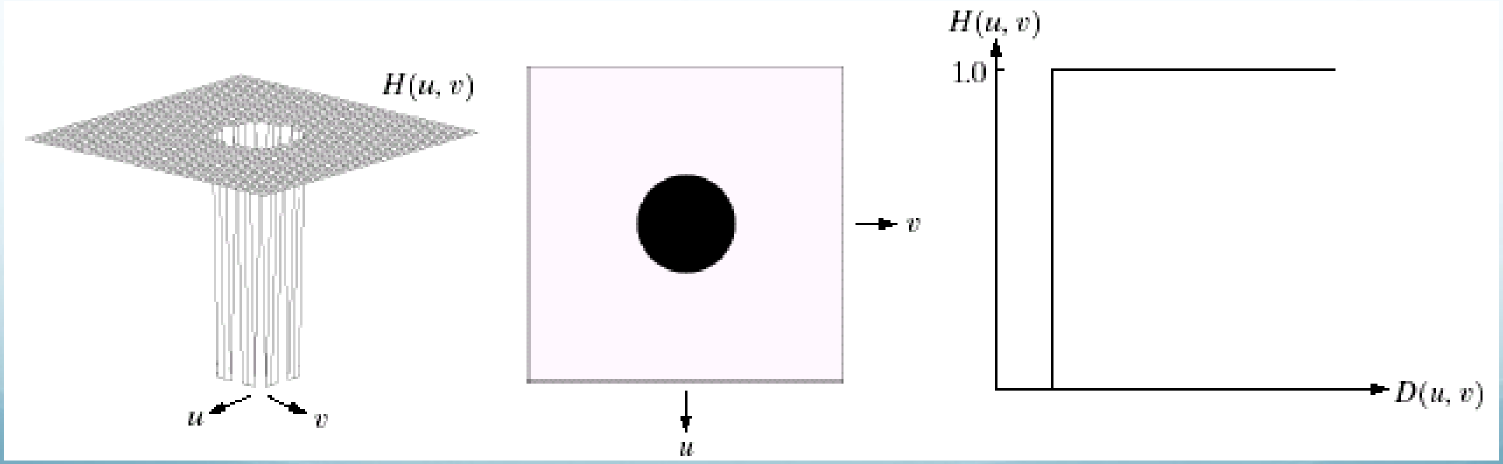
$$H_{hp}(u, v) = 1 - H_{lp}(u, v)$$

Ideal High Pass Filters

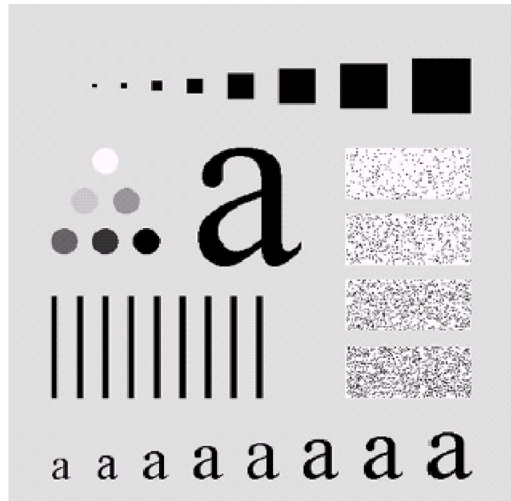
The ideal high pass filter is given as:

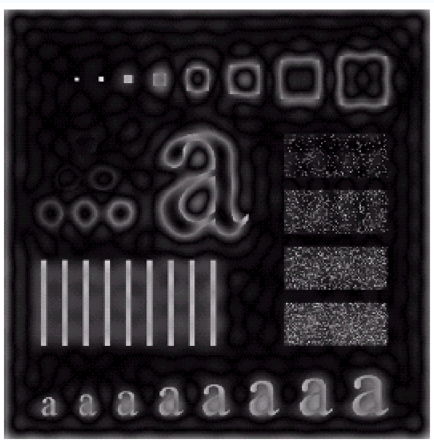
$$H(u,v) = \begin{cases} 0 \ \text{if} \ D(u,v) \leq D_0 \\ 1 \ \text{if} \ D(u,v) > D_0 \end{cases}$$

where D_0 is the cut off distance as before

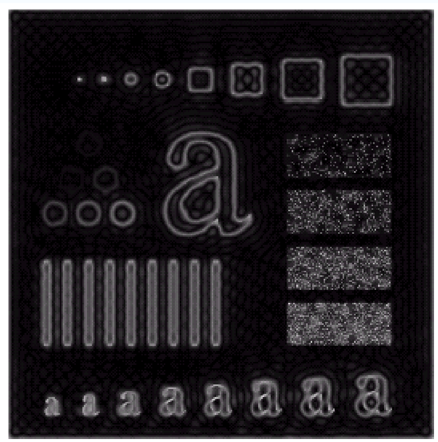


Ideal High Pass Filters

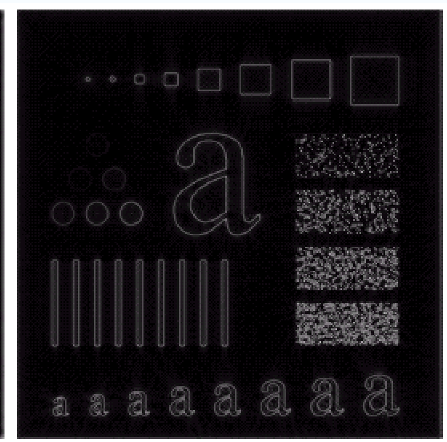




Results of ideal high pass filtering with $D_0 = 15$



Results of ideal high pass filtering with $D_0 = 30$



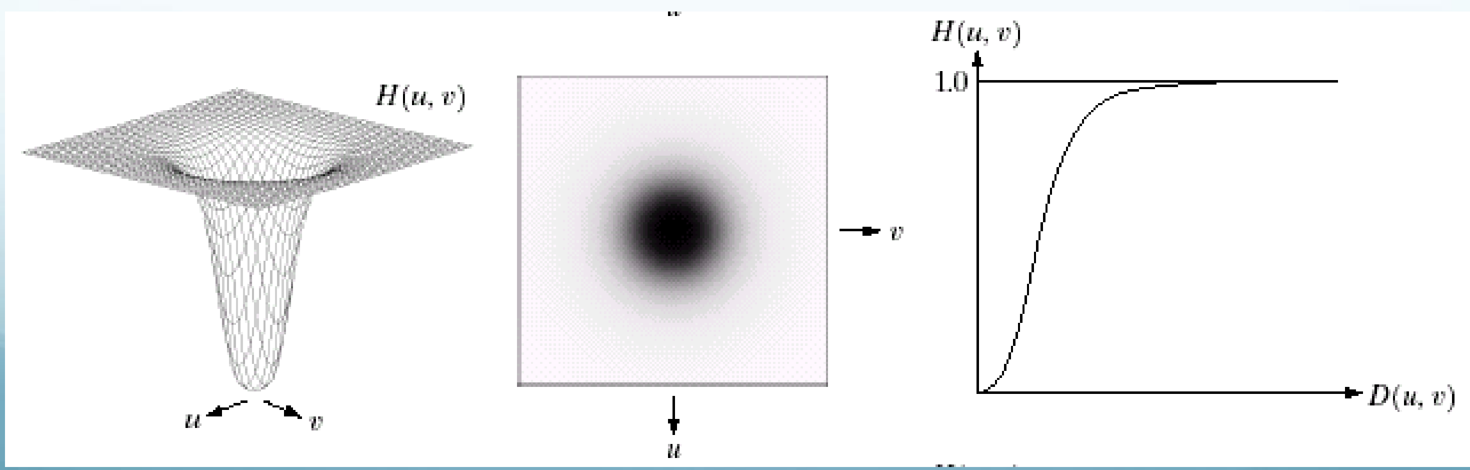
Results of ideal high pass filtering with $D_0 = 80$

Butterworth High Pass Filters

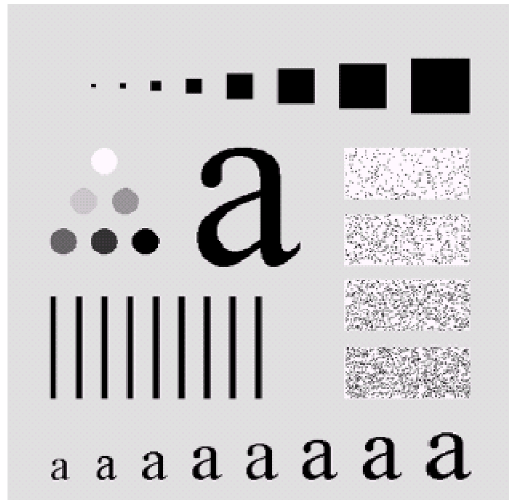
The Butterworth high pass filter is given as:

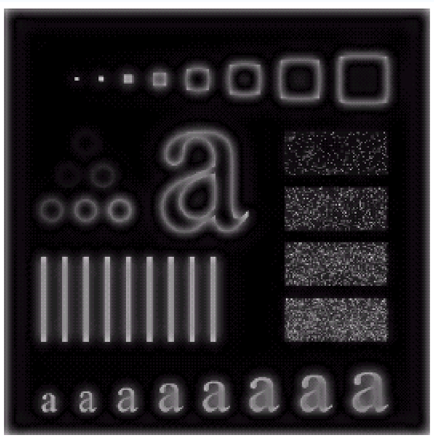
$$H(u,v) = \frac{1}{1+[D_0 / D(u,v)]^{2n}}$$

where n is the order and D_0 is the cut off distance as before

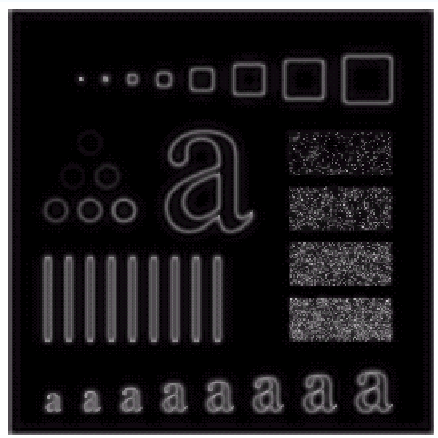


Butterworth High Pass Filters

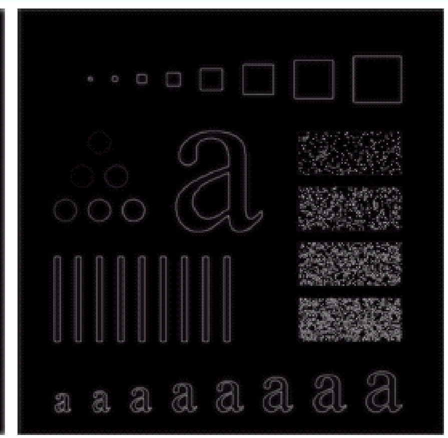




Result of Butterworth high pass filtering of order 2 with D_0 = 15



Result of Butterworth high pass filtering of order 2 with D_0 = 30



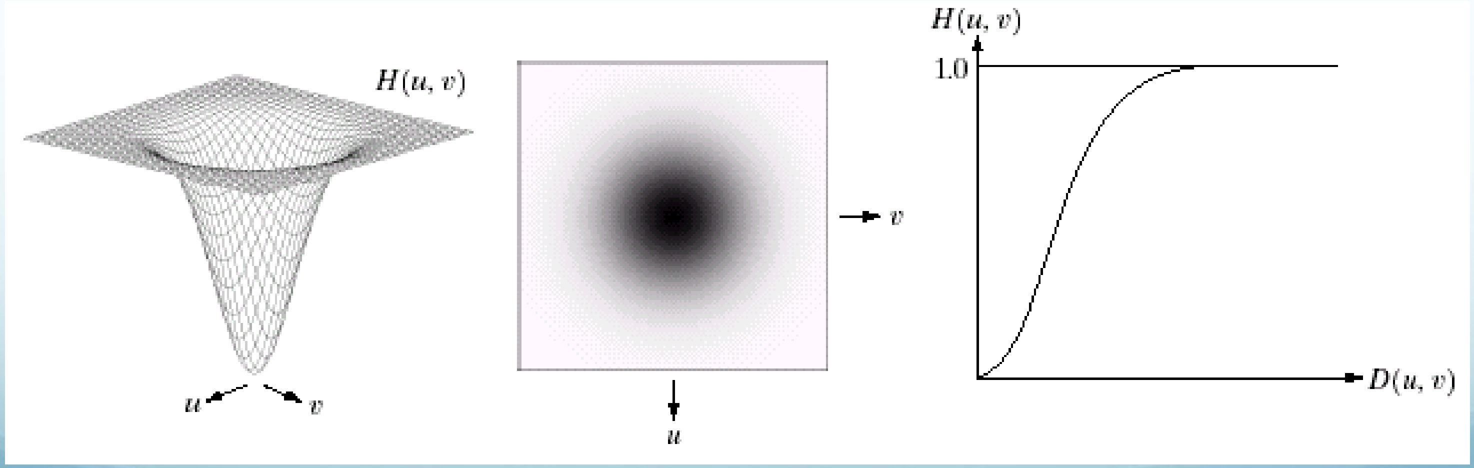
Result of Butterworth high pass filtering of order 2 with D_0 = 80

Gaussian High Pass Filters

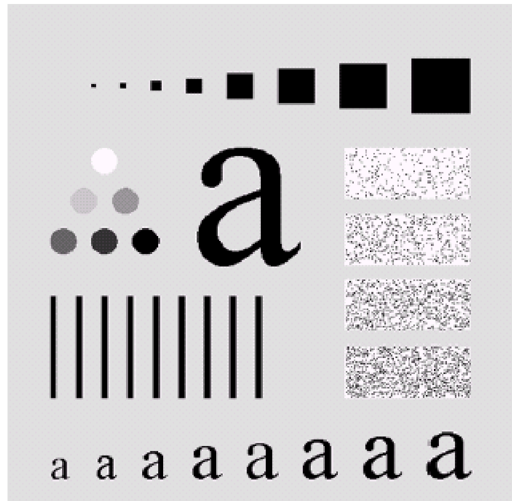
The Gaussian high pass filter is given as:

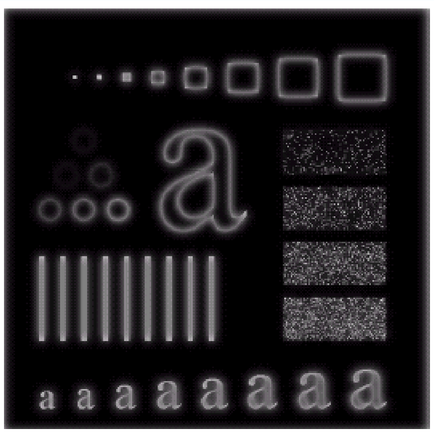
$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

where D_0 is the cut off distance as before

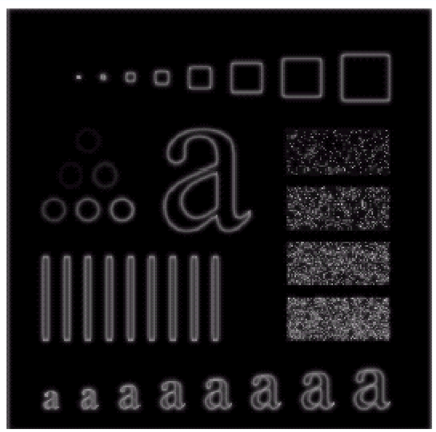


Gaussian High Pass Filters





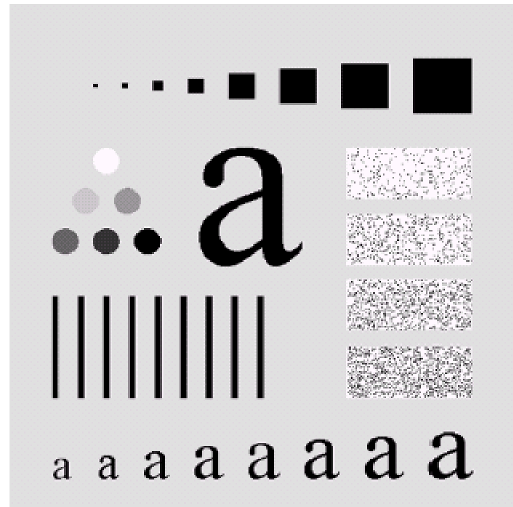
Result of Gaussian high pass filtering with D_0 = 15

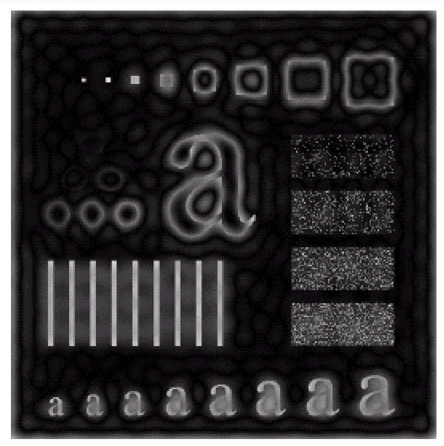


Result of Gaussian high pass filtering with D_0 = 30

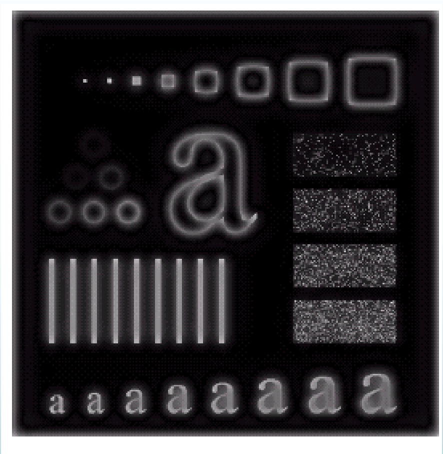
Result of Gaussian high pass filtering with D_0 = 80

Highpass Filter Comparison

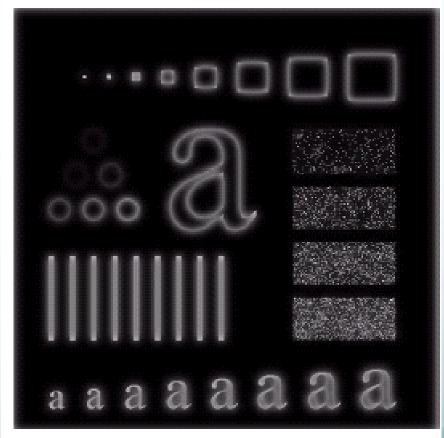




Results of ideal high pass filtering with D_0 = 15

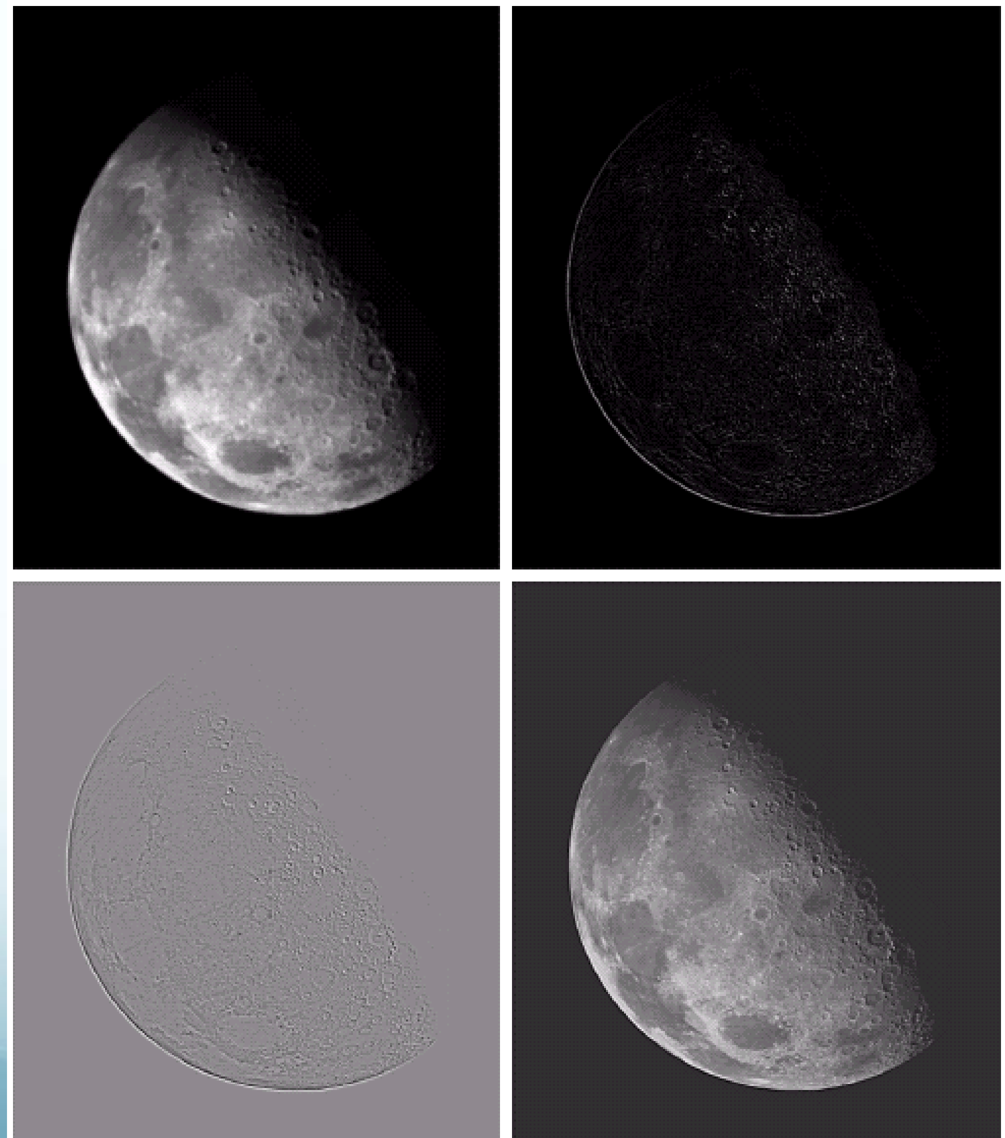


Results of Butterworth high pass filtering of order 2 with D_0 = 15



Results of Gaussian high pass filtering with D_0 = 15

Frequency Domain Laplacian Example



Original image

Laplacian filtered image

Laplacian image scaled

Enhanced image

Fast Fourier Transform

The reason that Fourier based techniques have become so popular is the development of the *Fast Fourier Transform (FFT)* algorithm

Allows the Fourier transform to be carried out in a reasonable amount of time

Reduces the amount of time required to perform a Discrete Fourier Transform by a factor of 100 – 600 times!

Frequency Domain Filtering & Spatial Domain Filtering

Similar jobs can be done in the spatial and frequency domains

Filtering in the spatial domain can be easier to understand

Filtering in the frequency domain can be much faster – especially for large images

Summary

We examined image enhancement in the frequency domain

- The Fourier series & the Fourier transform
- Image Processing in the frequency domain
  - Image smoothing
  - Image sharpening
- Fast Fourier Transform

Next time we will begin to see noise removal and color image processing